SOLUTION DESIGN · TECHNICAL ANNEX

Meridian Retail Group — Global Omnichannel Lakehouse

Full technical annex: current state, data inventory, target architecture, governance, delivery, commercials and risk.

CLIENT	DOCUMENT	DATE	VERSION	STATUS
Meridian Retail Group (fictional)	Solution design — technical annex	27 September 2026	1.0	For decision

Illustrative — fictional client; all figures illustrative. Every number in this document (revenue, volumes, timelines, benefits, effort and cost) exists to make the reasoning concrete and checkable. Nothing here is a quotation, a commitment, or a statement about any real organisation's estate.

Contents

Page numbers are the PDF page numbers of this document (A3 landscape), resolved by measuring the rendered output.

Meridian Retail Group — Global Omnichannel Lakehouse 4

How to read this document 5

1. Executive summary 6
- 1.1 The business problem, in five bullets 6
- 1.2 The proposed outcome 6
- 1.3 Headline benefits — order of magnitude 6
- 1.4 Timeline 7
- 1.5 The ask 7

2. Current state 8
- 2.1 The estate 8
- 2.2 The pain, in business terms 8
- 2.3 The pain, in engineering terms 8
- 2.4 What is genuinely good, and must be preserved 9

3. The data inventory 10
- 3.0 Inventory at a glance 10
- 3.1 SAP ECC — Finance, Inventory, Procurement, Own-brand Production 10
- 3.2 Salesforce — B2B wholesale, franchise and service 11
- 3.3 Point of Sale — three regional platforms 11
- 3.4 E-commerce clickstream 11
- 3.5 Product catalogue / PIM 12
- 3.6 Supply-chain EDI feeds 12
- 3.7 IoT — distribution centres and cold chain 13
- 3.8 Marketing platforms 13
- 3.9 Third-party market and reference data 14
- 3.10 HR — Workday 14
- 3.11 The legacy EDW 14
- 3.12 Reference and finance master data 15

4. Use cases and value 16
- UC-1 — Unified Customer 360 16
- UC-2 — Demand forecasting and replenishment 16
- UC-3 — Real-time inventory availability 16
- UC-4 — Markdown and promotional optimisation 17
- UC-5 — Supplier performance and claims recovery 17
- UC-6 — Personalisation and offer targeting 17
- UC-7 — Finance close acceleration 18
- UC-8 — Cold-chain compliance and waste reduction 18
- 4.9 Prioritisation matrix 18

5. Target architecture 20
- 5.1 The shape, in one paragraph 20
- 5.2 Zones and the medallion contract 20
- 5.3 Ingestion — the decision per source 20
- 5.4 Streaming versus batch, per use case 21
- 5.5 Unity Catalog design 22
- 5.6 The serving layer 23
- 5.7 Platform layer 24
- 5.8 Architecture decisions, consolidated 25

6. Data products and domain ownership 32
- 6.1 How the estate is carved 32
- 6.2 The domains 32
- 6.3 What a data product must publish 32
- 6.4 How ownership maps to the catalog 33

7. Governance, security and compliance 34
- 7.1 The obligation landscape 34
- 7.2 PII handling and pseudonymisation 34
- 7.3 Right to erasure — the mechanics 34
- 7.4 Residency 36
- 7.5 Access model for 400+ analysts 36
- 7.6 Sharing with suppliers and agencies 36
- 7.7 Audit and lineage evidence 37

8. Delivery plan 38
- 8.1 Shape of the programme 38
- 8.2 Phase 0 — Mobilise and prove (weeks 1–6) 38
- 8.3 Phase 1 — The thin slice (weeks 7–16), first value at week 10 38
- 8.4 Phase 2 — Core domains (weeks 17–40) 39
- 8.5 Phase 3 — Scale and migrate (weeks 41–72) 39
- 8.6 Migration approach — strangler, dual-run, decommission gates 39
- 8.7 What the client must provide 40

9. Team, ways of working and enablement 41

9.1 Pod shape 41

9.2 Rough FTE by phase (illustrative) 41

9.3 Client-side roles to be filled 42

9.4 CI/CD and environments 42

9.5 Testing and quality gates 42

9.6 Runbooks 43

9.7 Enablement and certification path 43

10. Cost model and commercials 45

10.1 The cost drivers 45

10.2 How we estimate 45

10.3 How we control cost 46

10.4 Illustrative effort and cost profile 46

10.5 The commercial construct we would propose 46

10.6 Payback logic 46

11. Risks and mitigations 48

12. Success metrics 49

12.1 Business KPIs 49

12.2 Platform SLOs 49

13. What we would draw next 51

14. Appendix 52

14.1 Glossary — client-facing terms 52

14.2 Assumptions register 52

14.3 Items marked to be confirmed 53

Meridian Retail Group — Global Omnichannel Lakehouse

Deep-dive business case · Case 01 of the Lakehouse Design Pattern series Prepared for: Group Executive Committee and the Data & Technology leadership team Version 1.0 · Status: for decision

Illustrative case — please read first

Meridian Retail Group is fictional. It is a composite drawn from engagements with large grocery, general-merchandise and own-brand CPG organisations. **Every number in this document — revenue, volumes, timelines, benefits, effort and cost — is illustrative** and exists to make the reasoning concrete and the arithmetic checkable. Nothing here is a quotation, a commitment or a statement of any real client's estate.

No Databricks list prices appear in this document. Commercials are expressed as *drivers, ratios and ranges*. Any real estate must be built from the client's own workload profile and the current Databricks and cloud price lists at the time of the proposal.

Where a platform behaviour or limit is material to the design and has not been re-verified against the current documentation, it is marked **[TBC against current docs]**.

How to read this document

This case is written for two readers at once. They should not read all of it.

If you are…	Read	Skip
The executive sponsor (CEO, CFO, COO, CDO)	§1 Executive summary, §4 Use cases and value, §8 Delivery plan, §10 Cost model, §11 Risks, §12 Success metrics	§3, §5, §6, §7 detail
The delivery architect / lead engineer	§2 Current state, §3 Data inventory, §5 Target architecture, §6 Data products, §7 Governance, §9 Team and ways of working, §13 Diagrams	§1 and §10 headline framing (useful context, not instruction)
The programme director / PMO	§8, §9, §10, §11, §12	§3, §5
Security, Legal, DPO	§3 (classification columns), §7 in full, §11	§4, §10

Two conventions run through the document:

1. **Executive sections contain no platform jargon.** Where a product name is unavoidable in an executive section it is glossed in §14. Architect sections are deliberately specific — a solution architect should be able to open §3 and §5 and start building without a further discovery cycle.
2. **Every architectural choice is written as a DECISION**, with the alternative that was rejected and the reason. A decision without a rejected alternative is not a decision; it is a preference. Challenge them: they are the point of the document.

This case is the first in a series. The flow diagrams listed in §13 will be drawn from it, so the naming, the zone boundaries and the per-source ingestion choices in §3 and §5 are written to be drawn literally.

1. Executive summary

1.1 The business problem, in five bullets

1. **Meridian cannot see its own customer.** A loyalty member who buys in a Lisbon store, browses on the app in Madrid and returns an item through the German call centre is three unrelated records. 40 million loyalty members are managed as 9 disconnected regional lists.
2. **Stock is in the wrong places.** Replenishment runs on a weekly, country-level forecast produced from data that is between 24 hours and 6 days old. On-shelf availability sits around 94.1% against a 97.0% sector benchmark; the gap is worth an estimated $160–265m of lost sales a year.
3. **The close takes nine days and the answer is contested.** Finance, Commercial and Supply Chain each reconcile from a different extract. Month-end consumes roughly 14 FTE of manual reconciliation and the Board sees trading numbers when they are already history.
4. **The platform is out of road.** The central warehouse runs at 91% of capacity, the overnight batch frequently lands after the 07:00 business deadline, and the analytics cluster is two major versions behind support. Growth in e-commerce and own-brand manufacturing has nowhere to go.
5. **The data team is firefighting, not building.** Approximately 70% of the 48-person data engineering capacity is consumed by incident response, re-runs and manual reconciliation. The BI backlog stands at 340 requests with a median age of 11 months.

1.2 The proposed outcome

A single governed data platform — a **lakehouse** — that holds Meridian's trading, customer, supply chain and finance data once, in one shape, with one set of access rules, and serves it to every consumer: dashboards, forecasting models, the replenishment engine, the app, suppliers and auditors.

The programme replaces the central warehouse and the ageing analytics cluster over 24 months using a **strangler** approach — new consumers are pointed at the new platform first, the old systems are drained rather than lifted, and each decommission is gated on a proven reconciliation. Nothing is switched off on faith.

1.3 Headline benefits — order of magnitude

These are **illustrative ranges**, expressed as annual run-rate at steady state (end of Year 3), with the assumption that drives each one. §4 carries the full working. They are a case for doing the work, not a business plan: each must be re-based on Meridian's own actuals in Phase 0.

Value driver	Mechanism	Annual run-rate (illustrative)	Confidence
Inventory reduction	Daily SKU/store forecast replaces weekly country forecast; safety stock re-cut	$50–75m P&L (on $270–410m working capital release)	Medium-high
On-shelf availability	Real-time inventory position and exception-driven replenishment	$45–75m gross margin	Medium
Markdown optimisation	Price and markdown decisions on daily elasticity rather than calendar rules	$55–110m gross margin	Medium
Personalisation and loyalty	Unified customer view drives offer targeting across 9 e-commerce markets	$25–45m gross margin	Medium-low
Supplier performance recovery	OTIF and quality claims evidenced from shared, agreed data	$15–30m	Medium-high
Waste and shrink (cold chain)	Sensor-driven intervention before an excursion becomes a write-off	$10–20m	Medium
Finance close and reporting effort	9-day close to 4-day close; reconciliation automated	$6–9m cost, plus decision speed	High
Legacy run-cost avoidance	Warehouse and cluster decommission; licence and hardware exit	$9–14m	High
Indicative total		**$215–378m**	

Programme investment (illustrative): $28–42m across 24 months, including services, platform consumption, dual-run and client-side backfill. Payback on the *credible lower half* of the benefit range falls inside Year 2. The logic is in §10.6.

1.4 Timeline

Phase	Weeks	What the business gets
0 — Mobilise and prove	1–6	Foundations live; access, security and network resolved; benefits re-based on actuals
1 — Thin slice	7–16	**First value at week 10**: daily store-level sales and stock position for one region, with a governed dashboard replacing four spreadsheets
2 — Core domains	17–40	Customer 360, unified sales, inventory and product; forecasting in pilot
3 — Scale and migrate	41–72	All 14 countries; warehouse consumers migrated; dual-run and reconciliation
4 — Decommission and hand over	73–96	Legacy warehouse and cluster off; client team running the platform

1.5 The ask

#	Decision requested	By
1	Approve Phase 0 and Phase 1 (weeks 1–16) at an illustrative **$3.4–4.3m** of services plus platform consumption, with a formal gate at week 16 before Phase 2 is committed.	ExCo
2	Name a single accountable **executive sponsor** with authority across Commercial, Supply Chain, Finance and IT. A steering committee is not a sponsor.	CEO
3	Mandate **SAP data access** for the programme, including the basis team's time. This is the single most common cause of delay in this class of programme (§11, R2).	CFO / CIO
4	Start the **network and connectivity work in week 1** (private connectivity, firewall rules, identity federation). Lead times here, not engineering, set the critical path (§11, R8).	CIO
5	Accept the **decommission gates in §8.6** as binding — including the commitment that the legacy warehouse is switched off, not left running "just in case". The benefit case depends on it.	CFO

2. Current state

2.1 The estate

Layer	System	Scale	Condition
Central warehouse	On-premises MPP appliance (Teradata-class)	480 TB, ~11,000 tables, ~2,900 registered reports	91% capacity; next hardware refresh due in 14 months; support renewal in 9
Analytics cluster	Hadoop (CDH-lineage), 120 nodes, HDFS + Hive + Spark 2.x	2.1 PB raw, ~60% cold	Two major versions behind support; no clear upgrade path; 3 of 5 original engineers have left
ETL	Legacy graphical ETL tool	43 point-to-point job streams, ~2,700 individual jobs	No version control on 60% of jobs; change lead time 6–8 weeks
ERP	SAP ECC 6.0, single global instance, 3 company-code groups	~8,500 active tables; finance, inventory, procurement, own-brand production	Stable; heavily customised; the extraction conversation is political
CRM	Salesforce (Sales + Service Cloud) — B2B wholesale and franchise	~4.2m records	Well run; API limits are the constraint
POS	Three regional POS platforms (EU, APAC, Americas)	2,400 stores	Different schemas, different close-of-day timing, different offline behaviour
Digital	Homegrown clickstream pipeline, Kafka to HDFS	~360m events/day	Built by a team that no longer exists; no schema registry discipline
Last mile	Excel and Google Sheets	~1,400 "critical" spreadsheets identified in a 2025 audit	This is where most decisions are actually made

2.2 The pain, in business terms

Month-end close takes nine working days. Days 1–4 are spent waiting for the overnight batch to produce a clean trial position; days 5–7 are reconciliation between the warehouse, SAP and the regional POS totals; days 8–9 are commentary. Roughly 14 FTE across Group and regional finance are consumed each month, and the Board's trading review discusses a month that closed nine days ago.

Stock-outs are structural, not incidental. Replenishment proposals are generated weekly from a country-level forecast. A store that sells out on a Tuesday is not corrected until the following Monday's run. Measured on-shelf availability is 94.1%; the top-500 SKUs sit at 96.3% and the long tail at 91.8%. Commercial estimates — and the programme should validate — that closing half the gap to the 97.0% benchmark is worth 0.6–1.0% of store revenue.

There is no single customer view. Loyalty is administered per country. A member's e-commerce identity is an email address; their store identity is a card number; their service identity is a Salesforce contact. Match rates between the three are unknown because nobody has ever computed them. Marketing spends an estimated $18–24m a year on audiences it cannot deduplicate.

The data team spends 70% of its time firefighting. The 48-person team logged 1,190 production incidents in the last 12 months. The top three causes are upstream schema changes arriving without notice, late-arriving POS files from stores with poor connectivity, and batch windows overrunning into business hours. None of these are hard problems; they are unmanaged ones.

Everything important ends in a spreadsheet. The trading pack, the markdown decision, the supplier scorecard and the availability review are all assembled by hand. The numbers in them do not always agree, and no two are assembled the same way twice.

2.3 The pain, in engineering terms

Symptom	What it actually means	Consequence
No lineage	Nobody can answer "what feeds this number" without reading ETL job definitions by hand	Impact analysis for any change takes 2–3 weeks; most changes are therefore not made
No tests	Fewer than 5% of pipelines have any automated data assertion; none have unit tests	Defects are found by business users, in production, at month-end
Overnight batch only	Every pipeline is a nightly full or delta extract; the window is 22:00–06:30 and frequently breaches	No use case needing data more recent than yesterday can be built at all
Cost opacity	On-premises capital cost is not attributed to workload; the Hadoop cluster has no chargeback	No one can say what any report costs, so nothing is ever retired
Point-to-point sprawl	43 job streams connect ~19 systems directly to ~11 consumers	Every new consumer adds N new integrations, not one
No environment discipline	Two environments (dev, prod); dev has 4% of prod data and a different schema on 200+ tables	"It worked in dev" is not evidence of anything
Schema drift is unmanaged	Source changes arrive as overnight failures	~11 incidents/month, each 2–6 hours to triage

Symptom	What it actually means	Consequence
No reprocessing capability	Fixing a bad day means a manual, hand-written backfill script	Corrections take days and sometimes introduce new errors

2.4 What is genuinely good, and must be preserved

Not everything must be replaced. The programme explicitly protects three things:

- **The SAP finance chart of accounts and the company-code hierarchy.** These are correct, governed and auditable. The lakehouse conforms to them; it does not reinvent them.
- **The Salesforce B2B process.** The wholesale team's way of working is effective. The platform reads Salesforce; it does not attempt to re-platform it.
- **The store-level merchandising hierarchy** maintained in the PIM. It is imperfect, but it is the one dimension the whole business already agrees on. It becomes the conformed product dimension.

3. The data inventory

This is the section the delivery architect builds from. Each source is characterised with the same attribute set so that the ingestion map (§13, diagram 2) can be drawn directly from it.

Reading the tables. *Change rate* is the volume of new or changed records per period, not the total. *Latency requirement* is the **business** requirement, established with the consuming use case — not the technically achievable figure. *Ingestion pattern* is the decision, justified in §5.3.

Classification key: **P0** = no personal data · **P1** = internal, no personal data, commercially sensitive · **P2** = personal data (GDPR Art. 4) · **P3** = special category or payment data.

3.0 Inventory at a glance

#	Source	Domain	Landing volume/day	Latency need	Class	Ingestion pattern
1	SAP ECC	Finance, Inventory, Procurement, Production	~45 GB	4 h (inventory), daily (finance)	P1/P2	CDC via managed database connector + gateway
2	Salesforce	B2B wholesale, franchise, service	~1.5 GB	1 h	P2	Lakeflow Connect managed SaaS connector
3	POS (3 platforms)	Sales transactions	~12 GB	15 min (two regions), 4 h (one)	P2/P3	Auto Loader on landing volumes; Kafka streaming for one region
4	E-commerce clickstream	Digital	~90 GB compressed	1–5 min	P2	Structured Streaming from Kafka
5	Product / PIM	Product	~0.4 GB	1 h	P0	Lakeflow Connect database connector
6	Supply-chain EDI	Supply chain	~2 GB	30 min	P1	Auto Loader on SFTP-landed files
7	IoT — DC and cold chain	Supply chain, Quality	~8 GB	1 min (alerting)	P0	Streaming ingest (Kafka / direct push)
8	Marketing platforms (6)	Marketing	~3 GB	Daily (hourly for suppression)	P2	Lakeflow Connect where available; Auto Loader otherwise
9	Third-party market data	Commercial	~1 GB	Weekly / daily	P1	OpenSharing where supported; COPY INTO otherwise
10	HR (Workday)	People	~0.1 GB	Daily	P2/P3	Lakeflow Connect managed SaaS connector
11	Legacy EDW	All (historical)	One-off, plus daily delta during dual-run	Daily	P1/P2	Lakehouse Federation for read; bulk export for history
12	Reference and finance master	Finance, Reference	<0.1 GB	Daily	P0	COPY INTO from a governed volume

Steady-state landing is therefore **~165 GB/day compressed**, or roughly **60 TB/year** of bronze growth before history migration. History migration adds an estimated **190 TB** (the EDW subset worth keeping, plus 5 years of POS and 3 years of clickstream from the Hadoop cluster). Sizing for the first three years should assume **~500 TB** of managed storage under a retention policy, not the 2.6 PB that exists today — a substantial part of the Hadoop estate is duplicate and expired data that should be left behind, not migrated. That triage is a Phase 0 deliverable.

3.1 SAP ECC — Finance, Inventory, Procurement, Own-brand Production

Attribute	Detail
Business domain	Finance (GL, AP, AR), Inventory and stock movements, Procurement, Production orders
What it holds	~8,500 active tables; **~350 in scope**. Core: BKPF/BSEG (documents), MARA/MARC/MBEW (material master and valuation), MSEG/MKPF (movements), EKKO/EKPO (purchasing), AUFK/AFKO (production orders), plus the company-code and profit-centre hierarchies
Format / protocol	Oracle-backed SAP ECC, on-premises. Extraction candidates: log-based CDC from the underlying database, SAP-native extractors (ODP/SLT), or a batch table export
Volume	~60 TB in the underlying database; in-scope subset ~14 TB
Change rate	~40m changed rows/day, concentrated in MSEG/MKPF (stock movements, ~26m/day) and BSEG (~6m/day). Month-end multiplies finance volumes by ~4x for two days
Latency requirement	**Inventory movements: 4 hours** (drives availability and replenishment). **Finance: daily, by 06:00 local** (drives the close). Master data: daily
Quality issues	Heavy custom fields (Z-fields) with no documentation; three company-code groups use different posting conventions; MBEW valuation is restated retrospectively at period close, so "yesterday's" figure can change; deleted-flag semantics differ by module
PII / regulatory	**P1** mostly. **P2** in vendor and employee-linked records (HR postings, expense). Finance data is in scope for statutory audit and SOX-equivalent controls
Ingestion pattern	**Log-based CDC through a managed database connector** (ingestion gateway plus ingestion pipeline, landing into streaming tables). See DECISION D-03

Attribute	Detail
Risks	The SAP basis team owns the decision and has historically resisted log-based CDC on the production database. Mitigation and escalation path in §11, R2. Licensing implications of third-party extraction **[TBC with the client's SAP account team]**

3.2 Salesforce — B2B wholesale, franchise and service

Attribute	Detail
Business domain	B2B wholesale accounts, franchise partners, service cases
What it holds	~4,200 wholesale accounts, ~19,000 contacts, ~230k opportunities (lifetime), ~1.4m service cases, plus custom objects for franchise agreements
Format / protocol	Salesforce REST/Bulk API, via a managed connector
Volume	~4.2m records total; ~1.5 GB/day of change payload
Change rate	~180k record changes/day; spikes around quarter-end
Latency requirement	**1 hour** for service cases (feeds the supplier and customer-issue view); daily is sufficient for accounts and opportunities
Quality issues	Free-text account naming; duplicate accounts across country orgs (estimated 6–9%); the franchise custom object is maintained by hand
PII / regulatory	**P2** — named contacts, email, phone, business address. GDPR applies
Ingestion pattern	**Lakeflow Connect managed SaaS connector** into Unity Catalog streaming tables. See §5.3, row "Salesforce"
Risks	The API call budget is shared with three other integrations. Incremental behaviour depends on source-side change tracking; confirm per-object coverage **[TBC against current connector docs]**

3.3 Point of Sale — three regional platforms

Attribute	Detail
Business domain	Sales transactions, tender, promotions applied, loyalty card capture
What it holds	Basket header and line, tender line, promotion line, void/refund, till and operator reference. ~4.0m baskets/day, ~23m line items/day across 2,400 stores
Format / protocol	**EU (1,050 stores):** close-of-day compressed JSON pushed to object storage, one file per store per day, plus intraday partial files every 15 min. **Americas (780 stores):** near-real-time publish to a managed Kafka topic. **APAC (570 stores):** nightly fixed-width flat files via SFTP
Volume	~12 GB/day compressed; ~4.4 TB/year. Peak trading (late December) is ~2.4x
Change rate	Append-only, but **late arrivals are routine**: ~1.5% of store-days arrive 1–3 days late (connectivity); a further ~0.2% arrive as corrections up to 14 days later
Latency requirement	**15 minutes** for the Americas and EU intraday feeds (real-time availability, fraud); **4 hours** post-close for APAC. Trading dashboards need the prior day complete by **06:00 local**
Quality issues	Three schemas and three basket-identity schemes; store clocks drift (observed up to 40 minutes); the EU platform re-sends a whole day's file when a single line is corrected, so **idempotency is mandatory**; tender detail is masked inconsistently across regions
PII / regulatory	**P2** (loyalty card number links to an identified person). **P3** where truncated PAN or tender tokens are present — these must be dropped or tokenised at the boundary and must never land in bronze in the clear. PCI scope must be actively kept *out* of the lakehouse
Ingestion pattern	**Auto Loader with file-event notification** on a Unity Catalog external location for EU and APAC; **Structured Streaming from Kafka** for the Americas. Idempotency on basket business key plus source file identity. See §5.3, rows "POS"
Risks	The three-schema problem is the largest single silver-layer modelling effort in the programme. Budget a dedicated conforming workstream, not a mapping spreadsheet

3.4 E-commerce clickstream

Attribute	Detail
Business domain	Digital experience, personalisation, merchandising analytics, attribution
What it holds	Page view, product view, search, add-to-basket, checkout step, order confirmation, app screen events, plus a consent event stream. 9 e-commerce markets
Format / protocol	JSON over **Kafka** (self-managed, three clusters, one per cloud region), ~40 topics, no enforced schema registry today
Volume	~360m events/day; ~14.3m sessions/day; ~430 GB/day raw, ~90 GB/day compressed. Sustained ~4,200 events/sec, peak ~22,000/sec, Black Friday ~3x peak
Change rate	Append-only. Event schemas change 2–4 times a month per squad, unannounced
Latency requirement	**1–5 minutes** for personalisation and on-site merchandising; **1 hour** for trading dashboards; daily for attribution modelling
Quality issues	No schema registry, so malformed payloads are common; bot traffic is ~19% of events and is not flagged; session stitching across devices is unreliable; consent state is captured on a *separate* topic and must be joined before any personal use
PII / regulatory	**P2** — device ID, IP address, hashed email, logged-in customer ID. GDPR and ePrivacy apply; **consent state governs downstream use**, which is an architectural constraint, not a policy note
Ingestion pattern	**Structured Streaming from Kafka into a bronze streaming table**, schema-on-read with a rescued-data column so malformed events are captured rather than dropped. See §5.3, row "Clickstream"
Risks	Introducing a schema registry is a client-side prerequisite with its own change programme. Until then, the rescued-data column is the contract. Kafka retention (currently 3 days) must be raised to survive a weekend outage — **7 days minimum** is the ask

3.5 Product catalogue / PIM

Attribute	Detail
Business domain	Product, merchandising hierarchy
What it holds	~1.4m active SKUs, ~9m SKU-by-locale combinations, the 5-level merchandising hierarchy, supplier linkage, pack and dimension data, allergen and regulatory attributes, digital-asset metadata
Format / protocol	Commercial PIM on PostgreSQL; a read replica is available
Volume	~900 GB total; ~0.4 GB/day of change
Change rate	~250k attribute changes/day; seasonal range changes create bursts of ~2m changes over 48 hours, four times a year
Latency requirement	**1 hour.** Product is a conformed dimension — a new SKU missing from the dimension breaks every downstream join
Quality issues	Locale records are incomplete for smaller markets (~14% of SKU-locale rows have no description); the hierarchy has 3,100 SKUs parked in a "to be classified" node; supplier linkage is one-to-one in the model but many-to-one in reality
PII / regulatory	**P0**. Allergen and regulatory attributes carry consumer-safety obligations — accuracy matters more than latency
Ingestion pattern	**Lakeflow Connect database connector (PostgreSQL) against the read replica.** See §5.3, row "PIM"
Risks	The "to be classified" node must be a **blocking data-quality gate** in the silver layer, not a warning. Downstream reporting silently misattributes those SKUs today

3.6 Supply-chain EDI feeds

Attribute	Detail
Business domain	Supply chain, procurement, logistics
What it holds	Purchase order acknowledgements (855), advance ship notices (856), invoices (810), inventory advice (846), functional acknowledgements (997), from ~1,200 trading partners
Format / protocol	EDIFACT and X12, delivered to an SFTP endpoint by a VAN; ~14,000 documents/day
Volume	~2 GB/day; small files, high count (~14k files/day)

Attribute	Detail
Change rate	Append-only; ~3% of documents are re-sent corrections
Latency requirement	**30 minutes** for ASNs (drives DC inbound planning and the store availability projection); daily for invoices
Quality issues	Partner-specific dialect variation is the norm; ~6% of documents fail strict parsing today and are handled manually by an operations team; date and unit-of-measure conventions differ by partner
PII / regulatory	**P1** — commercially sensitive supplier terms. No personal data other than incidental contact names
Ingestion pattern	**Auto Loader on the SFTP landing zone** (registered as a Unity Catalog external location), parsing in a bronze-to-silver declarative pipeline, with a quarantine table for parse failures. See §5.3, row "EDI"
Risks	High file count at small file size is the classic Auto Loader cost trap — file-event notification rather than directory listing, plus compaction on the bronze table, is mandatory here

3.7 IoT — distribution centres and cold chain

Attribute	Detail
Business domain	Supply chain, food quality, sustainability
What it holds	Temperature and humidity from 38 DCs (~14,000 sensors), 2,600 refrigerated trailers (GPS plus temperature), and ~28,800 in-store refrigerated cases. Plus DC conveyor and sortation telemetry
Format / protocol	MQTT to a gateway to Kafka for DC and store; cellular telemetry via a vendor API for trailers
Volume	~66m readings/day at 1-minute intervals; small payloads (~120 bytes); ~8 GB/day
Change rate	Append-only, constant rate; gaps during connectivity loss are themselves signal
Latency requirement	**1 minute for excursion alerting.** This is the only genuinely sub-minute requirement in the estate, and it exists for food-safety reasons. Analytical use is daily
Quality issues	Sensor drift and failure (~1.5% of sensors report implausible values at any time); trailer telemetry drops out in transit and backfills in bursts; DC sensors are named inconsistently across the 38 sites
PII / regulatory	**P0**. Subject to **food-safety record-keeping obligations** (HACCP-equivalent) — retention and immutability of the evidence trail matter more than query performance
Ingestion pattern	**Streaming ingest into a bronze streaming table**, with alerting computed on a separate low-latency stream rather than on the analytical path. See §5.3, row "IoT"
Risks	Do not let a food-safety alert depend on the analytical platform's availability. The alerting path must be independently survivable; the lakehouse is the **system of record and of evidence**, not the safety interlock

3.8 Marketing platforms

Attribute	Detail
Business domain	Marketing, customer engagement
What it holds	Campaign, send, open, click and conversion from the email/CRM platform; paid media spend and performance from three ad platforms; app push and in-app messaging; the consent and preference store
Format / protocol	Mixed: two platforms have managed connectors, two expose REST APIs with daily export, one delivers scheduled CSV to object storage, one requires a vendor-side export job
Volume	~3 GB/day combined
Change rate	Daily batch for spend and performance; near-continuous for send/open/click (~40m events/day at campaign peak)
Latency requirement	**Daily** for spend and attribution. **Hourly** for suppression lists — a suppression that arrives late is a compliance incident, not a reporting delay
Quality issues	Campaign taxonomy is inconsistent across markets; spend is reported in platform currency with platform-defined FX; identity keys differ per platform (hashed email, device ID, platform user ID)
PII / regulatory	**P2**, and this is the highest-risk personal-data domain in the estate because it is where personal data is *acted upon*. Consent and suppression state are load-bearing
Ingestion pattern	**Lakeflow Connect managed connectors where one exists; Auto Loader on delivered files otherwise.** The consent/preference store is treated as a first-class source, not a marketing attribute. See §5.3, row "Marketing"

Attribute	Detail
Risks	Resist bespoke API pollers. Every bespoke poller is a future incident. Where no connector exists, prefer a vendor-side scheduled export to a landing zone

3.9 Third-party market and reference data

Attribute	Detail
Business domain	Commercial, category management, planning
What it holds	Market share and category panel data (weekly), competitor pricing (daily, collected by a vendor), weather actuals and forecast (hourly, by postcode district), public-holiday and school-term calendars, FX rates
Format / protocol	Varies by vendor: one panel vendor supports open data sharing natively; competitor pricing arrives as daily CSV to object storage; weather is a REST API; FX arrives from Treasury as a daily file
Volume	~1 GB/day
Change rate	Weekly (panel), daily (pricing, weather, FX)
Latency requirement	**Daily.** Weather-forecast refresh is 4-hourly and materially improves the demand forecast in fresh categories
Quality issues	Panel data is restated retrospectively for up to 6 weeks; competitor pricing has ~8% match failure against Meridian SKUs; weather geography does not align to store catchments without a mapping table Meridian must own
PII / regulatory	**P1**. **Licence terms constrain redistribution** — panel data typically may not be shared outside the licensed entity, which is a grant-model constraint (§7.6), not a footnote
Ingestion pattern	**OpenSharing where the vendor supports it** (zero-copy, no pipeline to run); **COPY INTO from a governed volume** for the low-volume file feeds. See §5.3, row "Third-party data"

3.10 HR — Workday

Attribute	Detail
Business domain	People, store operations, labour planning
What it holds	Worker, position, organisation, cost centre, contract type, absence, and the store-roster interface. ~85,000 employees, ~210,000 historical workers
Format / protocol	Workday, via a managed connector (HCM and Reports)
Volume	~0.1 GB/day
Change rate	~9,000 changes/day; monthly payroll cycles create spikes
Latency requirement	**Daily.** Labour scheduling is a separate operational system; the lakehouse serves analytics and cost allocation
Quality issues	Cost-centre mapping to SAP profit centres is maintained in a spreadsheet; contractor populations are partially outside Workday
PII / regulatory	**P2**, with **P3** elements (absence reason can imply health data; some jurisdictions treat union membership as special category). **Works-council consultation is required in at least 5 of the 14 countries before any employee-level analysis.** Treat this as a dependency with a lead time
Ingestion pattern	**Lakeflow Connect managed SaaS connector**, into a dedicated restricted catalog with no default grants
Risks	HR data is the most likely source of a privacy incident and the least likely to deliver value in Year 1. Recommend ingesting **organisation and cost-centre structures only** in Phases 1–2, deferring worker-level data until the works-council process completes

3.11 The legacy EDW

Attribute	Detail
Business domain	All — this is where 20 years of business logic lives
What it holds	~11,000 tables, of which ~1,900 are referenced by a report in the last 12 months and ~400 weekly. ~2,900 registered reports, of which ~620 have been opened in the last 90 days
Format / protocol	On-premises MPP appliance; JDBC access; bulk export to object storage is possible but constrained by appliance egress throughput (~4 TB/day sustained without harming the batch window)
Volume	480 TB. The **worth-keeping subset is estimated at 60–90 TB** — conformed history that cannot be rebuilt from source
Change rate	The whole warehouse is rebuilt nightly by the legacy ETL; during dual-run it continues to do so
Latency requirement	During dual-run: daily, for reconciliation comparison
Quality issues	Business logic is embedded in ETL, not documented. Several critical measures exist in three variants with different filters. Slowly-changing dimension history is inconsistent — some dimensions are Type 1 where they should be Type 2, which means **some history is simply gone** and cannot be recovered
PII / regulatory	**P1/P2** mixed, and poorly mapped. A discovery pass is required before any of it is migrated (§11, R7)
Ingestion pattern	Two distinct mechanisms, deliberately: **Lakehouse Federation** for read-only governed access during migration and reconciliation (no copy, no pipeline, immediate), and **a one-off bulk export** of the worth-keeping history subset. See DECISION D-12 (§5.8)
Risks	The instinct to "migrate the warehouse" must be resisted. The programme migrates **consumers and business logic**, re-derived from source, and keeps only history that cannot be rebuilt. §8.6 makes this a gate

3.12 Reference and finance master data

Attribute	Detail
Business domain	Finance, Reference
What it holds	Store master (opening dates, formats, catchments, square footage), calendar (fiscal, retail 4-5-4, trading day), FX rate table, company-code and profit-centre hierarchy, country and tax reference
Format / protocol	A mix of SAP extracts, a Finance-owned SharePoint workbook, and three spreadsheets
Volume	<0.1 GB/day
Change rate	Low — tens of rows per day — but **disproportionately impactful**. A store-master error misstates a region
Latency requirement	Daily
Quality issues	This is the least-governed and most-depended-on data in the estate. Store master exists in four places with three different store-code formats
PII / regulatory	**P0**
Ingestion pattern	**COPY INTO from a governed Unity Catalog volume**, with the spreadsheet replaced in Phase 2 by a small governed entry application. See §5.3, row "Reference"
Risks	Do not automate the ingestion of an ungoverned spreadsheet and call it solved. Reference data needs an **owner and a maintenance process** before it needs a pipeline

4. Use cases and value

Eight use cases are carried into the plan. They were selected because each one is (a) owned by a named executive, (b) measurable against an existing KPI, and © dependent on data the programme is already ingesting for another reason. Use cases that fail any of those three tests are parked, not refused.

A note on latency. The column "freshness it truly needs" is the most commonly over-specified number in a programme like this. Every stakeholder asks for real time. Only two of these eight use cases genuinely require it; asking for it elsewhere converts a cheap batch pipeline into an expensive streaming one for no decision benefit. The figures below are the ones the programme will defend.

UC-1 — Unified Customer 360

Business question	Who is this customer across every channel and country, what have they bought, what have they returned, what have they been sent, and what have they consented to?
Consumers	Marketing (campaign and audience build), Customer Service, Loyalty, Digital product teams, Group Commercial
Freshness truly needed	**Daily** for the analytical profile; **within 1 hour** for consent and suppression state; **near-real-time** only for the subset of attributes used by on-site personalisation (UC-6)
Depends on	POS (3.3), clickstream (3.4), Salesforce (3.2), marketing platforms and consent store (3.8), loyalty master (within POS/SAP), reference (3.12)
Value driver	Removes duplicate audience spend; makes every other customer-facing use case possible. **It is the enabling asset, and its value is mostly realised through UC-4 and UC-6.**
Value range	$8–14m direct (deduplicated media spend and service efficiency), plus it is a hard dependency for $25–45m in UC-6
Key assumption	Identity resolution achieves a 70–80% match rate between store loyalty card and e-commerce account within the loyalty population. If the true rate is below 55%, UC-6's value halves. **Measure this in Phase 0 on a sample — do not assume it.**

UC-2 — Demand forecasting and replenishment

Business question	How many units of each SKU will each store sell tomorrow, and what should be ordered today?
Consumers	Supply Chain planning, the replenishment engine (as an automated consumer), store operations
Freshness truly needed	**Daily, completed by 04:00 local** so the order cut-off is met. Intraday is not required; the order is placed once
Depends on	POS (3.3), SAP inventory and purchasing (3.1), PIM (3.5), EDI ASN (3.6), weather and calendar (3.9), promotional calendar (marketing, 3.8)
Value driver	Forecast error reduction lets safety stock be re-cut. Inventory is ~$3.4bn; carrying cost ~18%
Value range	**$50–75m annual P&L**, on a working-capital release of $270–410m
Key assumption	Forecast accuracy (WMAPE at SKU-store-day) improves from ~38% to 26–30%, and Supply Chain actually re-cuts safety-stock parameters on the back of it. **The second half of that sentence is where these programmes fail: a better forecast that nobody acts on is worth zero.** Make the parameter re-cut an explicit, owned deliverable

UC-3 — Real-time inventory availability

Business question	What is actually on the shelf, in the back room, in the DC and in transit, right now, for this SKU at this store?
Consumers	E-commerce (click-and-collect promise), store colleagues (app), the availability exception process, customer service
Freshness truly needed	**15 minutes.** This is one of the two genuine near-real-time cases. A collect promise made on 24-hour-old stock is a broken promise
Depends on	POS intraday (3.3), SAP goods movements (3.1), EDI ASN (3.6), PIM (3.5)
Value driver	On-shelf availability, plus reduced failed click-and-collect orders (currently ~3.1% of collect orders substitute or cancel)
Value range	**$45–75m gross margin**, from recovering 0.6–1.0% of store revenue on a 28% gross margin, phased over Years 2–3
Key assumption	The stock position derived in the lakehouse is accurate enough to be trusted operationally. It will not be on day one — book stock accuracy is estimated at 88% at SKU-store level. **Phase 2 must include a stock-accuracy measurement and a correction loop, or this use case will be rejected by stores**

UC-4 — Markdown and promotional optimisation

Business question	For this SKU in this store, what price or markdown maximises margin over the remaining life of the stock?
Consumers	Commercial / category managers, Pricing, regional trading teams
Freshness truly needed	**Daily.** Markdown decisions are taken on a weekly cycle with daily exception review
Depends on	POS (3.3), SAP inventory and valuation (3.1), PIM (3.5), competitor pricing and panel data (3.9), UC-2 forecast output
Value driver	Markdown spend is estimated at ~5.6% of store revenue ($1.5bn). Sector experience suggests 3–6% improvement is achievable when decisions move from calendar rules to measured elasticity
Value range	**$55–110m gross margin**
Key assumption	Category managers accept a recommendation engine in the loop. Recommend a **recommend-and-override** design, not automation — adoption is the binding constraint, not model quality

UC-5 — Supplier performance and claims recovery

Business question	Which suppliers are delivering on time and in full, at what quality, and what are we owed?
Consumers	Procurement, Supply Chain, Finance (claims), and **the suppliers themselves**
Freshness truly needed	**Daily**, with a weekly published scorecard
Depends on	EDI (3.6), SAP purchasing and goods receipt (3.1), DC IoT and quality events (3.7), PIM supplier linkage (3.5)
Value driver	OTIF penalties and quality claims that are currently not raised because they cannot be evidenced
Value range	**$15–30m**, of which roughly half is claims recovery and half is behavioural improvement once suppliers see the same scorecard
Key assumption	Suppliers can be given governed read access to their own slice of the scorecard without a per-supplier integration project. This is the case that justifies OpenSharing (§7.6)

UC-6 — Personalisation and offer targeting

Business question	What should we show or offer this customer, now, given what they have bought, browsed and consented to?
Consumers	Digital product, Marketing, the app and website, the loyalty offer engine
Freshness truly needed	**Session-level (seconds) for on-site behavioural signals**; daily for the profile attributes. This is the second genuine real-time case, and only for a narrow feature set
Depends on	UC-1 Customer 360, clickstream (3.4), POS (3.3), PIM (3.5), consent store (3.8)
Value driver	Incremental revenue on loyalty-attributed sales, currently ~44% of store revenue
Value range	**$25–45m gross margin**
Key assumption	Incremental lift of 1.5–3% on targeted segments, proven by holdout testing. **Require a holdout on every campaign from day one.** Without holdouts the value is unprovable and the programme will be argued about rather than extended

UC-7 — Finance close acceleration

Business question	What did we earn, where, and can we prove it — by day 4 rather than day 9?
Consumers	Group Finance, regional controllers, Investor Relations, external audit
Freshness truly needed	**Daily by 06:00 local**, with a flash trading position available intraday
Depends on	SAP finance (3.1), POS (3.3), reference and FX (3.12), legacy EDW history (3.11) during dual-run
Value driver	Reconciliation effort, decision latency, and the audit-evidence burden
Value range	**$6–9m cost**, principally 14 FTE of monthly reconciliation reducing to 4–5, plus a materially cheaper audit cycle. The larger, unquantified benefit is that the Board argues about the business rather than about the numbers
Key assumption	Finance accepts lakehouse-derived figures as the reporting basis after a **3-month parallel run with zero unexplained variance**. Do not promise a shorter parallel

UC-8 — Cold-chain compliance and waste reduction

Business question	Was this product held within temperature specification along its whole journey, and where are we losing product to excursions?
Consumers	Quality and food safety, DC operations, store operations, external auditors and regulators
Freshness truly needed	**1 minute for alerting** (operational, on the streaming path); daily for analysis and evidence
Depends on	IoT (3.7), EDI ASN (3.6), SAP movements (3.1), PIM product attributes (3.5)
Value driver	Product written off after an excursion, plus the cost of manual compliance evidence gathering
Value range	**$10–20m**, plus a materially reduced regulatory risk that is not monetised here
Key assumption	Alerts reach a person who can act within the product's thermal tolerance. **An alert with no owner is a log line.** The operating-model change is bigger than the engineering

4.9 Prioritisation matrix

Scores are 1–5. **Value** is annual run-rate benefit. **Effort** is delivery effort (5 = least effort). **Data readiness** is how usable the required sources are today (5 = ready). **Priority score** = Value x (Effort + Readiness) / 2, rounded. This is a conversation starter for the sponsor, not an algorithm; the sequencing in §8 also reflects dependency order.

Use case	Value	Effort (5 = easy)	Data readiness	Priority score	Phase
UC-7 Finance close	2	4	4	8	1–2
UC-3 Real-time availability	5	2	3	13	2–3
UC-2 Demand forecasting	5	2	3	13	2–3
UC-4 Markdown optimisation	5	3	3	15	3
UC-5 Supplier performance	3	4	3	11	2
UC-1 Customer 360	3	3	2	8	2
UC-6 Personalisation	4	2	2	8	3–4
UC-8 Cold chain	2	3	4	7	3

How to read this. UC-4 scores highest but depends on UC-2's forecast, so it cannot go first. UC-7 scores modestly but is sequenced early because it is the **credibility purchase**: a finance close that lands on day 4 is the outcome that buys the programme its Phase 3 budget. UC-1 scores lower than its strategic importance because its data readiness is genuinely poor — that is a finding, not a reason to skip it.

Deliberately excluded from Phase 1–3 scope, with the reason:

Parked use case	Why not now
Store labour optimisation	Depends on worker-level HR data, which is blocked by works-council consultation (§3.10)
Assortment and space planning	Requires planogram data that is not currently captured digitally in 6 of 14 countries
Dynamic pricing (as opposed to markdown)	Regulatory and brand risk in grocery; a separate decision, not a data decision
Scope-3 emissions reporting	Real and coming, but the data standard is still moving; revisit at the Phase 3 gate
Generative assistants over the estate	Becomes cheap and safe **after** the governed semantic layer exists, not before. Sequence it into Phase 4, not Phase 1

5. Target architecture

5.1 The shape, in one paragraph

Every source lands **once**, in its original grain, into a **bronze** zone owned by the platform team. Domain teams transform bronze into **silver** — conformed, deduplicated, quality-gated business entities with stable keys. **Gold** holds the consumption-shaped products: dimensional models, metric definitions, feature tables and extracts. One catalog per environment, one schema per domain, one owner per schema. Consumption is through governed SQL, dashboards, a natural-language interface and model serving. Nothing reads a source system directly except an ingestion pipeline, and nothing writes to gold except a pipeline in version control.

5.2 Zones and the medallion contract

The medallion pattern is only useful if each layer has a *contract* — a statement of what a consumer may and may not assume. Without contracts, "silver" becomes "wherever the last engineer stopped".

	Bronze	Silver	Gold
Purpose	Faithful, replayable capture of the source	Conformed, quality-gated business entities	Consumption-shaped products
Grain	Source grain, unaltered	Business entity grain (one row per basket line, per customer, per SKU-location-day)	Consumption grain (fact/dimension, aggregate, feature vector)
Schema	Source schema plus ingestion metadata; permissive, with a rescued-data column	Explicit, declared, versioned	Explicit, declared, versioned
Transformation allowed	None beyond decoding and metadata enrichment	Typing, deduplication, conforming, key assignment, SCD handling, quality enforcement	Joins, aggregation, business rules, derived measures
Quality	Recorded, never enforced (nothing is rejected)	**Enforced.** Expectations drop or quarantine failing rows; violations are counted and alerted	Assumed clean; assertions verify invariants only
Personal data	Present, restricted to the platform team and the ingestion identity	Pseudonymised; direct identifiers isolated in a separate restricted schema	Personal data only where a use case justifies it, under attribute-based policy
Who may read	Platform team, domain data engineers	Domain team, downstream domain engineers	Analysts, applications, partners, agents
Who may write	Ingestion pipelines only	The owning domain's pipeline only	The owning domain's pipeline only
Retention	Source-defined, typically 13–25 months; IoT evidence longer per food-safety obligation	Full history for entities requiring it; SCD2 where the business needs as-was reporting	Driven by the consuming use case
Reprocessing	Replayable from bronze without touching the source	Rebuildable from bronze	Rebuildable from silver

The bronze rule that matters most: bronze must be sufficient to rebuild everything downstream without going back to the source system. In this estate, that rule is what makes it safe to take a cautious extraction approach from SAP — one careful extract, replayed many times, rather than repeated extraction pressure on a production ERP.

Per-domain variations from the standard contract:

Domain	Variation	Why
Finance	Silver is **append-only with a restatement flag**; no in-place correction	Audit evidence — the as-reported figure must remain recoverable
Cold chain	Bronze retention extended to meet the food-safety record-keeping period	Regulatory obligation; confirm the exact period per country **[TBC with Legal]**
Clickstream	Bronze carries a **consent-state join key**; silver is filtered by consent at the point of creation	Personal data may not reach a usable shape before consent is checked
Customer	Direct identifiers live in a single restricted schema; silver and gold carry only a pseudonymous key	Makes erasure tractable (§7.3)
POS	Silver conforms three source schemas to one; the mapping is a versioned, reviewed artefact, not code comments	Three-way conforming is the highest-risk logic in the estate

5.3 Ingestion — the decision per source

The decision rule is a ladder, applied in order: **use the most managed option that meets the requirement, and drop a rung only when it does not.** Dropping a rung buys control and costs engineering time forever after.

> **Counting convention.** §3 counts **12 source systems**; the table below has **13 rows** because POS is one system estate served by two mechanisms (Auto Loader for EU and APAC, Kafka for the Americas). Marketing and third-party data likewise each carry two mechanisms within a single row. Count systems in §3, count feeds here.

1. **Managed connector (Lakeflow Connect)** — a source with a supported connector and no unusual requirement.
2. **Lakeflow Declarative Pipelines over Auto Loader / COPY INTO** — files arriving in object storage.
3. **Structured Streaming** — message buses, or anything needing sub-minute latency or custom stateful logic.
4. **Lakehouse Federation** — read the source in place, when copying is not justified.
5. **OpenSharing** — the data is already in someone else's lakehouse and can be shared without copying.

Source	Mechanism	Why	Rejected alternative, and why
SAP ECC (3.1)	**Managed database connector with CDC** (ingestion gateway plus ingestion pipeline)	Log-based CDC gives 4-hour inventory latency without repeated full extracts against production; schema evolution and retries are handled by the connector rather than by us	**Nightly full-table export:** cannot meet 4 hours, and the export window already competes with the ERP batch. **SAP-native extractors (ODP/SLT):** viable and may be politically easier, but adds an SAP-side component the client must license, operate and change-manage; keep it as the fallback if basis refuses log access (§11, R2)
Salesforce (3.2)	**Lakeflow Connect managed SaaS connector**	Supported source, modest volume, incremental by default; zero pipeline code to maintain	**Bespoke Bulk API extraction:** we would own pagination, throttling, soft deletes and schema drift forever, for no benefit
POS — EU and APAC (3.3)	**Auto Loader with file-event notification**, into a declarative pipeline	Millions of files over time at high file count; directory listing does not scale and costs more per file discovered. Exactly-once file processing removes the duplicate-file problem created by the EU platform's whole-day re-sends	**COPY INTO:** appropriate for thousands of files, not millions, and re-sent files would need manual handling. Keep COPY INTO available for targeted re-loads of a specific store-day
POS — Americas (3.3)	**Structured Streaming from Kafka** into a bronze streaming table	The source is already a stream and the 15-minute requirement is real	**Land to files first, then Auto Loader:** adds a hop and 10+ minutes of latency for no gain when the bus already exists
Clickstream (3.4)	**Structured Streaming from Kafka**, one stream per topic family, rescued-data column enabled	22k events/sec peak with unannounced schema change; the rescued-data column converts a nightly outage into a data-quality metric	**A managed streaming connector:** reduces code, but we need custom sessionisation and bot-flagging close to ingest and per-topic checkpoint control. Re-evaluate at the Phase 3 gate **[TBC against current connector capabilities]**
PIM (3.5)	**Lakeflow Connect database connector (PostgreSQL)** against the read replica	Supported source, read replica already exists, 1-hour latency easily met	**Federation:** product is joined in almost every downstream query; reading it remotely on every query would push repeated load onto the PIM database and give no history
EDI (3.6)	**Auto Loader on the SFTP landing location**, parsing in a declarative pipeline, failures to quarantine	14k small files a day is exactly the file-count profile Auto Loader exists for; parsing belongs in our code because the dialect variation is Meridian-specific	**A commercial EDI translation product:** the client already owns the VAN relationship; adding a second translation layer duplicates cost. **COPY INTO:** file count is too high
IoT (3.7)	**Streaming ingest into bronze**, plus a **separate low-latency alerting stream**	Two different consumers with two different availability requirements should not share one pipeline	**One pipeline serving both:** couples a food-safety alert to analytical platform availability. Rejected on risk, not cost
Marketing (3.8)	**Managed connector where one exists; Auto Loader on vendor-delivered files otherwise**	Six platforms, six different integration surfaces; standardising on "connector or file drop" keeps the count of bespoke components at zero	**Bespoke REST pollers per platform:** six components to own, each breaking independently
Third-party data (3.9)	**OpenSharing** for the panel vendor that supports it; **COPY INTO** for the small file feeds	Sharing avoids a copy, avoids a pipeline, and the restated history simply appears. COPY INTO is right for thousands of small files a year with occasional targeted re-load	**Building a pipeline for shared data:** copying data that is already accessible adds cost, latency and a new failure mode. **Auto Loader for the file feeds:** correct but heavier than needed at this file count
HR (3.10)	**Lakeflow Connect managed SaaS connector**, restricted catalog	Supported source, low volume, tight access control matters more than throughput	**Ad-hoc report exports to a share:** how HR data leaks
Legacy EDW (3.11)	**Lakehouse Federation** for reconciliation reads; **one-off bulk export** for retained history	Federation is available in days, needs no pipeline, and is exactly right for the temporary reconciliation workload. The retained history is a one-time move, so it does not deserve a pipeline either	**Building CDC from the appliance:** weeks of work on a system with a known end date. **Migrating all 480 TB:** migrates the problem
Reference (3.12)	**COPY INTO from a governed volume**	Tens of small files, occasional re-load, human-in-the-loop upload	**Auto Loader:** an always-on stream for a file that changes weekly is operational overhead without benefit

`D-01` **Ladder-first ingestion policy.** Every new source must justify *not* using the most managed available mechanism, in writing, in the design review. **Rejected:** a single standard pattern for all sources (typically "everything via Kafka" or "everything via files"), which is simple to describe and expensive to run — it forces every source through infrastructure it does not need and hides per-source cost.

DECISION D-02 — Declarative pipelines as the default transformation framework. Bronze-to-silver and silver-to-gold transformations are built as **Lakeflow Spark Declarative Pipelines**, with expectations as the quality mechanism, orchestrated by **Lakeflow Jobs**. **Why:** dependency resolution, incremental processing, data-quality expectations and lineage come from the framework rather than from convention; a new engineer reads the transformation, not the plumbing. **Rejected:** hand-written notebook pipelines orchestrated by job dependencies — maximum flexibility, but it reproduces exactly the situation described in §2.3, where every pipeline is a bespoke artefact and quality is whatever each author remembered to add.

DECISION D-03 — SAP extraction via log-based CDC, with an ODP/SLT fallback held in reserve. Why: it is the only option that meets the 4-hour inventory requirement without adding load to the ERP batch window. **Rejected:** nightly export (too slow), and SAP-native extraction as the primary route (adds a licensed, client-operated component and a change-control dependency). The fallback is documented and estimated now, in Phase 0, so that a basis-team refusal costs a re-plan and not a re-architecture.

5.4 Streaming versus batch, per use case

Streaming is chosen where a decision is made faster than daily. Everywhere else, scheduled incremental batch is chosen because it is cheaper to run, easier to reason about and much easier to reprocess.

Use case	Mode	Trigger	Justification
UC-3 Real-time availability	Streaming	Continuous on the POS Americas and IoT paths; 5-minute micro-batch for the EU intraday path	The decision (promise a collect slot) is made in seconds
UC-8 Cold-chain alerting	Streaming	Continuous	Food safety
UC-6 Personalisation (behavioural features)	Streaming	Continuous, narrow feature set only	Session relevance decays in minutes
UC-1 Customer 360	Batch	Scheduled incremental, several times daily; hourly for consent	Profiles do not change in seconds; consent does matter within the hour
UC-2 Demand forecasting	Batch	Daily, completed by 04:00	One order cut-off per day
UC-4 Markdown	Batch	Daily	Weekly decision cycle
UC-5 Supplier performance	Batch	Daily; weekly publication	Weekly scorecard
UC-7 Finance close	Batch	Daily by 06:00, plus an intraday flash	The ledger is not a stream

D-04 **Streaming pipelines run on a scheduled available-now trigger unless a use case demands continuous.** Only the three continuous cases above run always-on. Everything else that *reads* a stream runs on a schedule, processing all newly available data and stopping. **Why:** it gives streaming semantics (exactly-once, checkpointed, incremental, replayable) at batch cost, and it is a single configuration change to promote a pipeline to continuous if the business later needs it. **Rejected:** always-on streaming for everything — conceptually tidy, and it bills for idle time on pipelines whose consumers look at a dashboard once a morning.

5.5 Unity Catalog design

D-05 **Catalog per environment, schema per domain. Not catalog per domain.** Three catalogs: `mrg_dev`, `mrg_tst`, `mrg_prd`, each containing a schema per domain-and-layer. **Why:** the environment boundary is the one that must be absolutely impermeable — a development job must be structurally incapable of writing to production. Making environment the top-level securable makes that a property of the model rather than a matter of discipline. It also makes promotion a matter of substituting the catalog name, which keeps pipeline code identical across environments. **Rejected:** catalog-per-domain (`customer`, `sales`, `supply_chain`), which reads beautifully in a data-mesh diagram and then requires either a separate metastore per environment or a naming convention like `customer_prd` — at which point environment is in the name anyway, but is no longer enforced by the hierarchy. Also **rejected:** a single catalog with environment-prefixed schemas, which puts the most important boundary in a string.

Schema naming: `<layer>_<domain>`, lower snake case, no abbreviations that are not already enterprise vocabulary.

```
mrg_prd
├── bronze_sap              bronze_pos            bronze_clickstream
├── bronze_salesforce       bronze_pim            bronze_edi
├── bronze_iot              bronze_marketing      bronze_market_data
├── bronze_hr               bronze_reference
├── silver_sales            silver_inventory      silver_customer
├── silver_product          silver_supply_chain   silver_finance
├── silver_marketing        silver_people         silver_reference
├── silver_customer_pii     (restricted: direct identifiers only)
├── gold_trading            gold_customer_analytics
├── gold_supply_chain       gold_finance_reporting
├── gold_merchandising      gold_supplier_portal   (the OpenSharing surface)
├── ml_features             ml_models
└── ops_platform            (audit extracts, quality results, cost attribution, run metadata)
```

Volumes. Managed volumes for anything the platform itself writes (checkpoints in their own locations, quarantine payloads, export staging). External volumes for the three landing zones the client's systems write into: `landing_pos`, `landing_edi`, `landing_vendor_files`. A volume is the governed alternative to the mount paths and shared buckets that made the current estate unauditable.

Groups. Identity is federated from the corporate directory to the account and synchronised; groups are **account groups**, never workspace-local groups, so that any policy expression evaluates identically everywhere. The group taxonomy is deliberately small:

Group pattern	Example	Purpose
`mrg-platform-admin`	—	Metastore and platform administration; a small, named, audited set
`mrg-<domain>-owner`	`mrg-sales-owner`	Owns the domain's schemas; a **group**, never an individual
`mrg-<domain>-engineer`	`mrg-supply-chain-engineer`	Writes the domain's pipelines
`mrg-<domain>-reader`	`mrg-finance-reader`	Reads the domain's gold products
`mrg-analyst-<region>`	`mrg-analyst-emea`	Regional analyst population; the residency filter hangs off this
`mrg-pii-approved-<domain>`	`mrg-pii-approved-marketing`	Explicit, time-bound, approved access to unmasked personal data
`mrg-sp-<pipeline>`	`mrg-sp-ingest-sap`	Service principals; one per pipeline family, never shared

Grants model — four rules, applied without exception.

1. Grants are made **to groups, never to users**. An exception request becomes a group with one member and an expiry date, so it is visible.
2. Grants are made at **schema level** where possible, so new tables in a domain inherit access rather than requiring a new grant. Table-level grants are the exception, and each one should be explainable.
3. **Ownership is held by groups** — `mrg-<domain>-owner` — so that a leaver does not orphan a schema.
4. All grants are declared in configuration and applied by the deployment pipeline. **A grant made by hand in the UI is a defect**, and the weekly drift report treats it as one.

Tags. Governed tags, defined at account level, carry the classification that drives policy:

Tag key	Values	What it drives
`data_classification`	`p0`, `p1`, `p2`, `p3`	Masking policy, sharing eligibility, retention
`pii_type`	`email`, `phone`, `name`, `address`, `card_ref`, `loyalty_id`, `device_id`	Which masking function applies
`data_residency`	`eu`, `uk`, `apac`, `us`, `global`	Row-level residency filtering and share eligibility
`domain`	`sales`, `customer`, `supply_chain`, …	Ownership, cost attribution, catalogue navigation
`product_tier`	`certified`, `curated`, `experimental`	Consumer expectation and SLO tier (§6.3)
`cost_centre`	client cost-centre code	Chargeback

Note for the implementer: **column tags do not inherit from the parent table** and must be applied to the column directly. This makes column tagging a pipeline responsibility, not a one-off exercise — the declarative pipeline that creates a table also asserts its column tags.

`D-06` **Attribute-based access control on tags, not per-table masks.** Masking and row filtering are implemented as **ABAC policies attached at schema level, matching on governed tags** (for example: mask any column tagged `pii_type = email` for everyone except `mrg-pii-approved-*`). **Why:** it scales. With ~2,000 gold and silver tables and 400+ analysts, attaching a mask to each table individually is a permanent backlog and a guaranteed gap — the table created next Tuesday is unprotected until someone remembers. Tag-driven policy means a correctly tagged new table is protected the moment it exists. **Rejected:** per-table row filters and column masks — they remain the right tool for a handful of genuine one-offs, and they are the fallback for anything ABAC cannot express, but they cannot be the primary mechanism at this scale. **Implementation constraints to respect:** ABAC requires serverless compute or a sufficiently recent runtime on classic compute; older runtimes cannot read ABAC-secured tables at all. Materialised-view and streaming-table refreshes evaluate policies as the **pipeline owner**, so a pipeline owner with broad access can permanently materialise unmasked data — pipeline service principals must therefore be scoped as narrowly as analysts, not exempted. Quotas on policies per schema and per table are documented and generous but finite; the policy count here should stay in the low hundreds by design. Verify the current compute requirements and quota figures at build time **[TBC against current docs]**.

5.6 The serving layer

Consumer	Served by	Design note
~400 analysts, ad-hoc SQL	**Databricks SQL warehouses**, serverless, sized per workload class	Three warehouse classes: `xs-adhoc` (auto-stop aggressive), `m-bi` (dashboard refresh), `l-finance` (close period, scheduled scale-up). Analysts never choose their own compute
Executive and operational dashboards	**AI/BI dashboards** on gold and metric views	Dashboards read gold or metric views only — never silver. A dashboard that needs silver is a missing gold product
Consistent business definitions	**Metric views** in Unity Catalog	Sales, margin, availability and markdown are defined once, in one place, and consumed identically by SQL, dashboards, the natural-language interface and external BI. This is what finally kills the "three versions of gross margin" problem
Self-service questions from non-analysts	**AI/BI Genie**, scoped per domain over certified gold products	Genie spaces are scoped to a curated set of certified tables with documented columns. An ungoverned Genie space over the whole catalog produces confident wrong answers, which is worse than no answer
Data science and ML	Notebooks, **feature tables in Unity Catalog**, MLflow	Feature tables are governed objects in `ml_features`, not project-local artefacts. Training and serving read the same definition
Operational model consumption (UC-2, UC-4, UC-6)	**Model serving endpoints**	The replenishment engine and personalisation call an endpoint; they do not read tables. This keeps the operational contract narrow and versionable

Consumer	Served by	Design note
Existing enterprise BI tool	Direct connection to SQL warehouses over the governed layer	Existing investment is preserved; governance is not bypassed, because the warehouse enforces the same policies
Suppliers and agencies	**OpenSharing** from `gold_supplier_portal`	§7.6
Operational applications needing low-latency lookups	A serving store fed from gold	Do not serve a store-colleague app by querying the analytical warehouse on every request. The pattern and product choice here should be confirmed against the current platform options **[TBC against current docs]**

D-07 **Serverless SQL warehouses as the default, with classic compute only where a specific constraint requires it. Why:** an analyst population of 400 with bursty, unpredictable demand is the worst possible fit for provisioned capacity — it is either queuing or idling. Fast start-up removes the incentive for analysts to keep a warehouse running "just in case", which is the single largest source of waste we see in comparable estates. **Rejected:** provisioned classic warehouses sized to peak, which is cheaper per unit of work and much more expensive per unit of *useful* work, and which reintroduces capacity planning as a standing activity. Revisit for any workload found to be steady, predictable and large.

5.7 Platform layer

Workspaces. The workspace is a working environment, not a security boundary — Unity Catalog is the security boundary. Workspaces are therefore allocated for **operational and residency reasons only**.

Workspace	Region	Purpose
`mrg-eu-prd`	EU	Production workloads for EU/UK data; the primary production workspace
`mrg-eu-nonprd`	EU	Dev and test; bound to `mrg_dev` and `mrg_tst` catalogs
`mrg-apac-prd`	APAC	Production for APAC-resident data where residency requires in-region processing
`mrg-amer-prd`	Americas	Production for Americas data
`mrg-eu-sandbox`	EU	Analyst and data-science exploration; separate cost centre, separate budget, aggressive policies

D-08 **Three regional production workspaces, one metastore per region, one shared governance model.** **Why:** data residency in 14 countries cannot be met by a single-region deployment, and processing location — not just storage location — is what several regulators care about. A metastore is regional, so multi-region necessarily means multiple metastores; cross-region access is then provided by OpenSharing rather than by direct grants. **Rejected:** a single global workspace and metastore (simplest to operate, fails the residency test), and a workspace per country (14 workspaces, 14 sets of policies, and an operations team that spends its life on drift). Confirm the current cross-metastore access options at design time **[TBC against current docs]**.

Compute strategy.

Workload	Compute	Reason
Scheduled pipelines and jobs	Serverless job compute	No cluster management, no idle, start-up cost is amortised over the run
Always-on streaming (3 pipelines)	Dedicated job compute, right-sized, autoscaling within narrow bounds	Continuous workloads are the one case where a persistent cluster is efficient
SQL and BI	Serverless SQL warehouses (§5.6)	Bursty demand
Data science exploration	Serverless notebook compute under a policy, in the sandbox workspace	Governed by policy and budget, not by asking people to be careful
Ad-hoc heavy engineering (migration backfills)	Job compute under a policy, spot workers with an on-demand driver, hard auto-termination	Interruptible work at a lower cost, without risking the driver

Cost controls — five, from the outset, not retrofitted.

1. **Enforced tagging** via compute policies for classic compute and usage policies for serverless, so that every unit of consumption carries `cost_centre`, `domain` and `environment`. Untagged consumption is unattributable and therefore unmanageable; a policy that permits it is a bug.

2. **Budgets with alert thresholds** per domain and per environment, reviewed weekly in the first six months and monthly thereafter.
3. **Attribution reporting** built in Phase 0 from the billing and pricing system tables, joined correctly (a temporal range join on the price validity window — a plain equality join on the SKU fans out across historical price versions and silently inflates every figure).
4. **Aggressive auto-stop** everywhere, plus a standing report of the ten most expensive queries, pipelines and dashboards each week, with a named owner for each.
5. **Storage lifecycle and retention** applied per zone, with automated table maintenance enabled so that optimisation and file cleanup are not a task someone forgets.

> D-09 **Cost attribution is built in Phase 0, before the first production pipeline. Why:** cost opacity is one of the named failures of the current estate (§2.3). A platform that cannot answer "what does this report cost" reproduces the problem in a new place, and retrofitting tags across hundreds of jobs is far more expensive than enforcing them from job one. **Rejected:** deferring cost management to Phase 3 "once there is something to measure", which is the normal approach and the reason the conversation is always uncomfortable.

5.8 Architecture decisions, consolidated

ID	Decision	Rejected alternative
D-01	Ladder-first ingestion: most managed mechanism that meets the need	One uniform pattern for all sources
D-02	Declarative pipelines as the default transformation framework, orchestrated by Lakeflow Jobs	Hand-written notebook pipelines
D-03	SAP via log-based CDC, ODP/SLT held as a costed fallback	Nightly full export; SAP-native extraction as primary
D-04	Scheduled available-now triggers unless a use case demands continuous	Always-on streaming everywhere
D-05	Catalog per environment, schema per domain	Catalog per domain; single catalog with prefixes
D-06	ABAC policies on governed tags for masking and row filtering	Per-table masks and filters as the primary mechanism
D-07	Serverless SQL warehouses by default	Provisioned classic warehouses sized to peak
D-08	Three regional production workspaces, regional metastores, cross-region via sharing	Single global workspace; workspace per country
D-09	Cost attribution built in Phase 0	Cost management deferred to Phase 3
D-10	Infrastructure and workloads deployed as **Declarative Automation Bundles** from Git (§9.4)	UI-created jobs promoted by export/import
D-11	Data products published as governed **schemas with contracts** (§6)	A central team owning all transformation
D-12	Legacy EDW read by federation during migration; only unrecoverable history is copied	Full warehouse migration
D-13	Pseudonymisation at the bronze-to-silver boundary (§7.2)	Full-estate masking policies over identifier-bearing tables
D-14	Supplier exchange via OpenSharing (§7.6)	A supplier portal; 1,200 per-supplier extracts

D-10 is stated in full in §9.4, D-11 in §6.1, D-13 in §7.2 and D-14 in §7.6, alongside the material they govern.

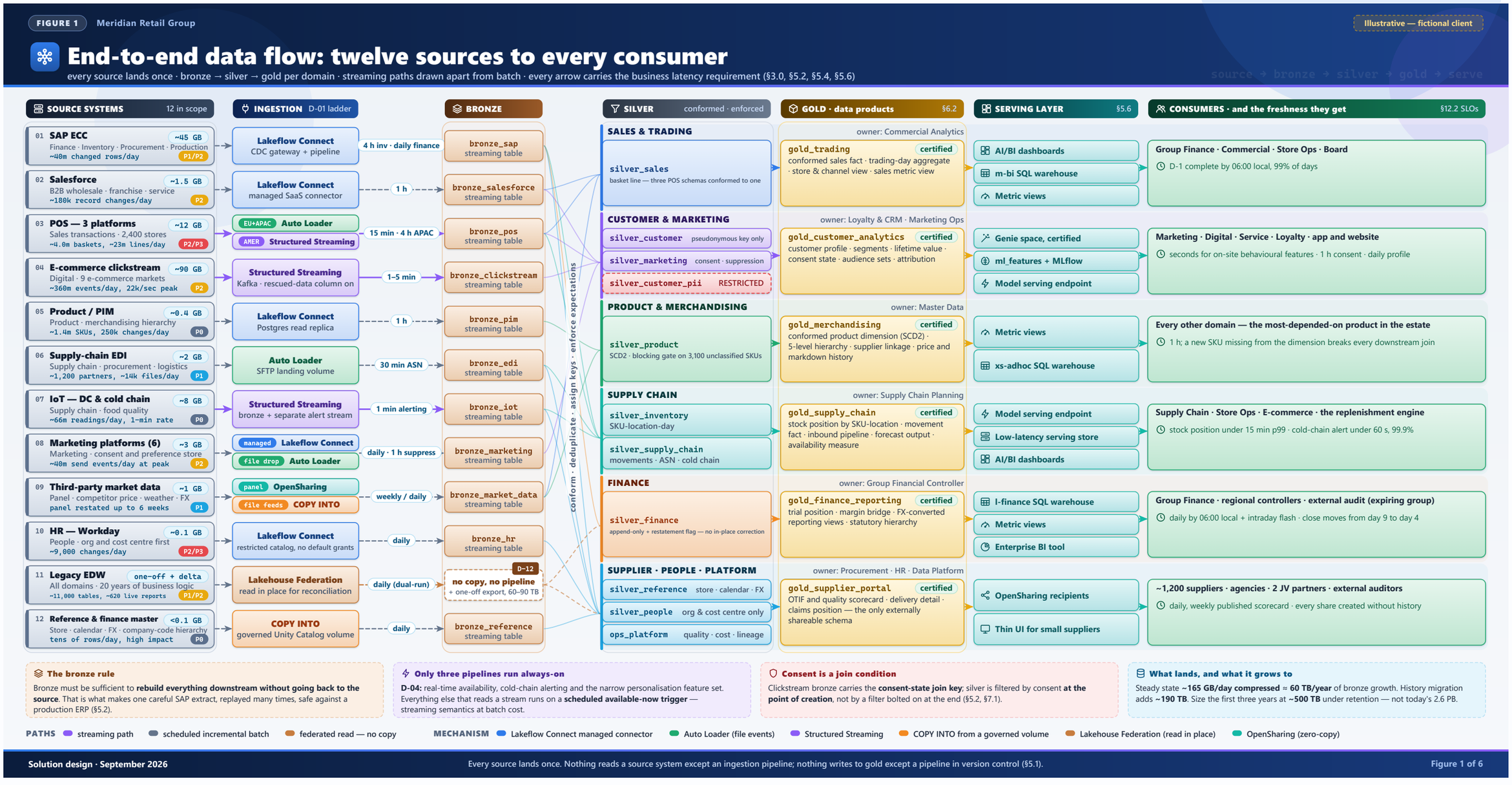


Figure 1 — End-to-end flow

`case01_fig1_end_to_end_flow.svg`

FIGURE 2 Meridian Retail Group

Illustrative — fictional client

Per-source ingestion map: the architect's build sheet

one row per source · what arrives · the factor that decides · the mechanism · what it lands as in Unity Catalog · and the alternative that was rejected (§3.1–3.12, §5.3)

D-01 · THE INGESTION LADDER

use the most managed option that meets the requirement — drop a rung only when it does not

1. **Managed connector** — Lakeflow Connect — a supported source with no unusual requirement — 4 sources
2. **Declarative pipeline over files** — Auto Loader / COPY INTO — files arriving in object storage — 5 feeds
3. **Structured Streaming** — Kafka and message buses — sub-minute latency or custom stateful logic — 3 sources
4. **Lakehouse Federation** — read the source in place — copying is not justified — temporary or read-only — 1 source
5. **OpenSharing** — zero-copy inbound share — the data is already in someone else's lakehouse — 1 vendor

D-01 · the policy, not a preference

Every new source must justify **not** using the most managed available mechanism, **in writing, in the design review**. Dropping a rung buys control and costs engineering time forever after. **Rejected:** a single standard pattern for all sources — simple to describe, expensive to run, and it hides per-source cost.

D-03 · SAP, with a fallback already costed

Log-based CDC is the only option that meets the **4-hour inventory** requirement without adding load to the ERP batch window. The **ODP/SLT fallback is designed and estimated in Phase 0**, so a basis-team refusal (risk R2) costs a re-plan, not a re-architecture.

D-12 · the warehouse is not migrated

Federation for reconciliation reads; a **one-off bulk export of the 60–90 TB that cannot be rebuilt from source**. The programme migrates **consumers and business logic**, re-derived from source. **Rejected:** migrating all 480 TB — that migrates the problem.

SOURCE	WHAT ARRIVES · protocol and shape	THE DECIDING FACTOR	MECHANISM	LANDS AS · Unity Catalog object	VOLUME	LATENCY NEED	REJECTED ALTERNATIVE — and why
SAP ECC §3.1 D-03	Oracle-backed SAP ECC, on-premises · ~350 of ~8,500 tables in scope: BKPF/BSEG, MARA/MARC/MBEW, MSEG/MKPF, EKKO/EKPO, AUFK/AFKO	Log-based CDC gives 4-hour inventory latency **without repeated full extracts against production**; schema evolution and retries are handled by the connector, not by us	**Managed database connector, CDC** — ingestion gateway + ingestion pipeline	`bronze_sap · streaming tables`	~45 GB/day	4 h inventory daily 06:00 finance	**Nightly full-table export:** cannot meet 4 hours, and the export window already competes with the ERP batch. **SAP-native ODP/SLT as primary:** viable, politically easier, but adds a licensed client-operated component — held as the costed fallback.
Salesforce §3.2	Salesforce REST / Bulk API · ~4.2m records, ~180k changes/day, spikes at quarter-end	Supported source, modest volume, **incremental by default** — zero pipeline code to maintain	**Lakeflow Connect SaaS connector** — managed, serverless	`bronze_salesforce · streaming tables`	~1.5 GB/day	1 h service cases daily accounts	**Bespoke Bulk API extraction:** we would own pagination, throttling, soft deletes and schema drift forever, for no benefit. Confirm per-object incremental coverage **[TBC against current connector docs]**.
POS — EU + APAC §3.3	EU (1,050 stores): close-of-day compressed JSON to object storage, plus 15-min intraday partials. APAC (570 stores): nightly fixed-width flat files via SFTP	Millions of files over time at high file count — **directory listing does not scale**. Exactly-once file processing removes the duplicate created by the EU platform's whole-day re-sends	**Auto Loader, file-event notification** — into a declarative pipeline	`external volume landing_pos → bronze_pos`	~12 GB/day (all 3 POS)	15 min EU intraday 4 h post-close APAC	**COPY INTO:** appropriate for thousands of files, not millions, and re-sent files would need manual handling. Keep COPY INTO available for a targeted re-load of one store-day.
POS — Americas §3.3	780 stores publishing near-real-time to a managed Kafka topic	The source **is already a stream** and the 15-minute requirement is real	**Structured Streaming from Kafka** — bronze streaming table	`bronze_pos · streaming table idempotent on basket key + file id`	–	15 min	**Land to files first, then Auto Loader:** adds a hop and 10+ minutes of latency for no gain when the bus already exists.
E-commerce clickstream §3.4	JSON over self-managed Kafka · 3 clusters, ~40 topics, no enforced schema registry · consent captured on a separate topic	22k events/sec peak with **schema change 2–4 times a month, unannounced**; the rescued-data column turns a nightly outage into a data-quality metric	**Structured Streaming from Kafka** — one stream per topic family, rescued data on	`bronze_clickstream · streaming tables + consent-state join key`	~90 GB/day (430 GB raw)	1–5 min personalisation 1 h dashboards	**A managed streaming connector:** less code, but we need custom sessionisation, bot-flagging (~19% of events) and per-topic checkpoint control close to ingest. Re-evaluate at the Phase 3 gate **[TBC]**.
Product / PIM §3.5	Commercial PIM on PostgreSQL · a read replica is available · ~1.4m SKUs, ~9m SKU-locale rows	Supported source, **the read replica already exists**, 1-hour latency easily met	**Lakeflow Connect database connector** — PostgreSQL, against the read replica	`bronze_pim · streaming tables`	~0.4 GB/day	1 h	**Federation:** product is joined in almost every downstream query — reading it remotely on every query would push repeated load onto the PIM database and give no history.
Supply-chain EDI §3.6	EDIFACT and X12 delivered to an SFTP endpoint by a VAN · 855 / 856 / 810 / 846 / 997 from ~1,200 trading partners	14k small files a day is **exactly the file-count profile Auto Loader exists for**; parsing belongs in our code because the dialect variation is Meridian-specific	**Auto Loader on the SFTP landing** — declarative parsing, failures to quarantine	`external volume landing_edi → bronze_edi + quarantine table`	~2 GB/day ~14k files	30 min ASN daily invoices	**A commercial EDI translation product:** the client already owns the VAN relationship; a second translation layer duplicates cost. **COPY INTO:** file count far too high. File-event notification plus bronze compaction is mandatory here.
IoT — DC & cold chain §3.7	MQTT → gateway → Kafka for 38 DCs and 28,800 store cases · cellular telemetry via a vendor API for 2,600 trailers	Two consumers with **two different availability requirements** should not share one pipeline	**Streaming ingest into bronze** — plus a separate low-latency alerting stream	`bronze_iot · streaming table extended retention (food safety)`	~8 GB/day ~66m readings	1 min alerting daily analysis	**One pipeline serving both:** couples a food-safety alert to analytical platform availability. **Rejected on risk, not cost** — the lakehouse is the system of record and of evidence, not the safety interlock.
Marketing platforms (6) §3.8	Two platforms have managed connectors, two expose REST with daily export, one delivers scheduled CSV to object storage, one needs a vendor-side export job	Six platforms, six integration surfaces. Standardising on **"connector or file drop" keeps the count of bespoke components at zero**	**Connector where one exists** — Auto Loader on vendor-delivered files otherwise	`bronze_marketing · consent and preference store as a first-class source`	~3 GB/day	daily spend 1 h suppression lists	**Bespoke REST pollers per platform:** six components to own, each breaking independently. Where no connector exists, prefer a **vendor-side scheduled export** into a landing zone.
Third-party market data §3.9	One panel vendor supports open data sharing natively · competitor pricing as daily CSV · weather over REST · FX as a daily Treasury file	Sharing **avoids a copy, avoids a pipeline, and the restated history simply appears**. COPY INTO is right for thousands of small files a year with occasional targeted re-load	**OpenSharing where supported** — COPY INTO for the small file feeds	`shared catalog — no copy landing_vendor_files → bronze_market_data`	~1 GB/day	weekly panel daily price / FX	**Building a pipeline for shared data:** copying what is already accessible adds cost, latency and a new failure mode. **Auto Loader for the file feeds:** correct but heavier than needed. Licence forbids redistribution — an explicit exclusion from every outbound share.
HR — Workday §3.10	Workday HCM and Reports via a managed connector · ~85,000 employees, ~210,000 historical workers	Supported source, low volume — **tight access control matters more than throughput**	**Lakeflow Connect SaaS connector** — into a restricted catalog	`bronze_hr · restricted no default grants`	~0.1 GB/day	daily	**Ad-hoc report exports to a share:** how HR data leaks. Ingest **organisation and cost-centre structure only** in Phases 1–2; worker-level data waits for works-council consultation in 5+ countries.
Legacy EDW §3.11 D-12	On-premises MPP appliance over JDBC · bulk export constrained to ~4 TB/day sustained without harming the batch window	Federation is **available in days, needs no pipeline**, and is exactly right for a temporary reconciliation workload on a system with a known end date	**Lakehouse Federation for reads** — one-off bulk export for retained history	`foreign catalog — no copy + one-off load of 60–90 TB history`	one-off + daily delta	daily (dual-run only)	**Building CDC from the appliance:** weeks of work on a system with a known end date. **Migrating all 480 TB:** migrates the problem. Only history that cannot be rebuilt from source is copied.
Reference & finance master §3.12	A mix of SAP extracts, a Finance-owned SharePoint workbook and three spreadsheets · store master exists in four places with three code formats	Tens of small files, occasional re-load, **human in the loop** on the upload	**COPY INTO from a governed volume** — spreadsheet replaced by an entry app in Phase 2	`managed volume → bronze_reference`	<0.1 GB/day	daily	**Auto Loader:** an always-on stream for a file that changes weekly is operational overhead without benefit. Do not automate an ungoverned spreadsheet and call it solved — reference data needs an **owner and a maintenance process** before it needs a pipeline.

MECHANISM COLOUR — Lakeflow Connect managed connector · Auto Loader (file-event notification) · Structured Streaming · COPY INTO from a governed volume · Lakehouse Federation · OpenSharing

Every mechanism named here is already in the case — no product is invented for the drawing.

Solution design · September 2026

Five mechanisms, thirteen feeds, zero bespoke pollers. Each drop down the ladder is written down and defended in the design review (D-01).

Figure 2 of 6

Figure 2 — Per-source ingestion map

`case01_fig2_ingestion_map.svg`

FIGURE 3 Meridian Retail Group

Illustrative — fictional client

Unity Catalog object model: catalogs, schemas, groups and policy

one catalog per environment, one schema per domain-and-layer · grants to groups at schema level · governed tags drive ABAC, not per-table masks (§5.5, §6.4 · D-05, D-06)

mrg_dev · mrg_tst · mrg_prd

THE OBJECT TREE

§5.5 · schema naming: <layer>_<domain>, lower snake case

mrg_dev
development mode · prefixed resources
paused schedules · masked representative subset

mrg_tst
deployed automatically on merge to main
identical shape, not a sample

mrg_prd (expanded below)
production mode · service principal only
Git branch validation · no cluster overrides

D-05: the environment boundary is the one that must be absolutely impermeable — a development job must be **structurally incapable** of writing to production. The three catalogs hold the **identical schema set**, so promotion is a catalog-name substitution and pipeline code is unchanged.

BRONZE · one schema per source · platform team writes, ingestion pipelines only 11

bronze_sap	bronze_pos	bronze_clickstream	bronze_salesforce
bronze_pim	bronze_edi	bronze_iot	bronze_marketing
bronze_market_data	bronze_hr	bronze_reference	

SILVER · one schema per domain · the owning domain's pipeline writes 10

silver_sales	silver_inventory	silver_customer	silver_product
silver_supply_chain	silver_finance	silver_marketing	silver_people
silver_reference	silver_customer_pii		

GOLD · the published data products · analysts, applications, partners, agents read 6

gold_trading	gold_customer_analytics	gold_supply_chain	gold_finance_reporting
gold_merchandising	gold_supplier_portal		

ML AND PLATFORM · governed, not project-local 3

ml_features	ml_models	ops_platform

VOLUMES · the governed alternative to mount paths and shared buckets

MANAGED
- checkpoints (own locations)
- quarantine payloads
- export staging

EXTERNAL · client systems write here
- landing_pos
- landing_edi
- landing_vendor_files

External volumes sit on **Unity Catalog external locations**, which is also what keeps lineage intact — path-based access and renamed objects break it (§7.7).

GROUPS, OWNERSHIP AND GRANTS

§5.5, §6.4

GROUP TAXONOMY — account groups, never workspace-local

GROUP PATTERN	EXAMPLE	PURPOSE
mrg-platform-admin	–	metastore admin
mrg-<domain>-owner	mrg-sales-owner	owns the schemas
mrg-<domain>-engineer	mrg-supply-chain-eng	writes pipelines
mrg-<domain>-reader	mrg-finance-reader	reads gold
mrg-analyst-<region>	mrg-analyst-emea	residency filter
mrg-pii-approved-<domain>	mrg-pii-approved-mktg	expiring, justified
mrg-sp-<pipeline>	mrg-sp-ingest-sap	one per pipeline

FOUR GRANT RULES — applied without exception

1. **Groups, never users**
   an exception becomes a group with one member **and an expiry date**, so it is visible
2. **Schema level, not table level**
   new tables in a domain **inherit** access; every table-level grant must be explainable
3. **Ownership is held by groups**
   ALTER SCHEMA … OWNER TO mrg-<domain>-owner — a leaver never orphans a schema
4. **Declared in configuration**
   applied by the deployment pipeline. **A grant made by hand in the UI is a defect**, and the weekly drift report treats it as one

HOW A GRANT FLOWS — and the cross-domain rule

mrg-finance-reader (account group) → mrg_prd (USE CATALOG) → gold_finance_reporting (USE SCHEMA + SELECT) → every table in it (inherits)

- Another domain's **gold** — read freely
- Another domain's **silver** — explicit, recorded agreement — it couples release cycles
- Another domain's **bronze** — prohibited

GOVERNED TAGS AND ABAC

§5.5 · D-06 · §7.2

GOVERNED TAGS — defined at account level

TAG KEY	VALUES	WHAT IT DRIVES
data_classification	p0 p1 p2 p3	masking, sharing, retention
pii_type	email phone name address	which masking function
	card_ref loyalty_id device_id	
data_residency	eu uk apac us global	row filter, share eligibility
domain	sales customer supply_chain	ownership, cost, navigation
product_tier	certified curated experimental	consumer expectation, SLO
cost_centre	client cost-centre code	chargeback

D-06 · WHERE POLICY ATTACHES

- CATALOG mrg_prd
  - SCHEMA gold_customer_analytics — ABAC policy attaches HERE
    - TABLE customer_profile — protected the moment it exists
      - COLUMN email_address — tag applied by the pipeline that creates it

Column tags do not inherit from the parent table. Tagging is therefore a pipeline responsibility: the declarative pipeline that creates a table also asserts its column tags.

⚠ Implementation constraints to respect

ABAC needs **serverless compute or a sufficiently recent runtime**; older classic runtimes cannot read ABAC-secured tables at all. Materialised-view and streaming-table refreshes evaluate policy **as the pipeline owner**, so pipeline service principals must be scoped as narrowly as analysts. Keep the policy count in the low hundreds by design. **[TBC against current docs]**

WHERE PERSONAL DATA LIVES

- **bronze_* (all sources)** — identifiers present in the source payload — platform team and ingestion identities only
- **silver_customer_pii** — the only schema holding direct identifiers · 2 service principals + 1 approval group
- **silver_* / gold_*** — pseudonymous surrogate key only; masked by ABAC where an identifier must appear
- **gold_supplier_portal** — no personal data at all — the only externally shared schema

D-05 · catalog per environment, schema per domain — the rejected alternatives

Catalog-per-domain (customer, sales, supply_chain) reads beautifully in a data-mesh diagram and then requires either a separate metastore per environment or a naming convention like customer_prd — at which point environment is in the name anyway, but is no longer enforced by the hierarchy. Also rejected: **a single catalog with environment-prefixed schemas**, which puts the most important boundary in a string.

D-06 · ABAC on tags, not per-table masks — why it scales

With **~2,000 gold and silver tables and 400+ analysts**, attaching a mask to each table individually is a permanent backlog and a guaranteed gap — the table created next Tuesday is unprotected until someone remembers. Tag-driven policy means a correctly tagged new table is protected the moment it exists. **Per-table row filters and column masks remain the fallback** for genuine one-offs ABAC cannot express.

Solution design · September 2026 — The workspace is a working environment; Unity Catalog is the security boundary. Workspaces are allocated for operational and residency reasons only (§5.7). — Figure 3 of 6

Figure 3 — Unity Catalog object model

case01_fig3_unity_catalog_model.svg

FIGURE 4 Meridian Retail Group

Illustrative — fictional client

Environments and promotion: repository to production

everything deployed as bundles from Git · one commit promoted, never rebuilt · the identity changes at each step · gates are automated, not meetings (§9.4 D-10, §9.5, §5.7 D-08)

dev → tst → prd

ONE COMMIT, PROMOTED — never rebuilt per environment

D-10 · Declarative Automation Bundles from Git

GIT MONOREPO
the single source of truth

one directory per domain, shared libraries in **common/**

trunk-based: short-lived feature branches off **main**

jobs, pipelines, schemas, volumes, dashboards, monitors **and grants** all declared here

identity: the engineer
`no deployment`

MERGE GATE
- unit tests
- contract tests
- peer review

must pass to merge

dev
target: mrg_dev

development mode — **prefixed resources**, per-developer isolation

schedules paused; nothing runs on a timer

data: a representative **masked subset**, not 4% of prod with a different schema

identity: the developer
`databricks bundle deploy -t dev`

TEST GATE
- data-quality expectations
- reconciliation
- freshness

automatic on merge

tst
target: mrg_tst

deployed **automatically on merge to main**

identical in shape to production — not a sample

reconciliation and quality suites run here before anything is promoted

identity: CI service principal
`workload identity federation`

RELEASE GATE
- performance regression
- grant-model validation

manual approval

prd
target: mrg_prd

production mode — **service principal identity**

Git branch validation; no cluster overrides permitted

a **gated action from the same commit** that passed test — never a rebuild

identity: prd service principal
`manual approval, same commit`

WEEKLY DRIFT JOB · compares deployed state against the repository and against the declared grant model

drift is a defect with an owner

WHAT THE BUNDLE DEPLOYS

§9.4

DEPLOYED BY THE BUNDLE
- Jobs (Lakeflow Jobs)
- Pipelines (declarative)
- Schemas
- Volumes
- Dashboards
- Quality monitors
- **Grants**

NOT DEPLOYED — and why
- **Data.** Dev holds a representative masked subset — producing it is a Phase 0 deliverable
- **Anything created in the UI.** A hand-made job or grant is a defect, not a shortcut
- **Cluster overrides in prd.** Production mode forbids them
- **Long-lived tokens in CI.** Deployment uses workload identity federation
- **The legacy ETL's undocumented logic.** It is re-derived, reviewed and committed — not lifted

D-10 · the rejected alternative

Creating jobs in the UI and promoting by export/import is faster on day one, and it guarantees environment drift, undocumented production changes, and a platform nobody can rebuild. The current estate's inability to answer "what changed" (§2.3) is the root cause of most of its incidents.

WHO MAY DEPLOY WHAT

PATH	WHO	TRIGGER
feature branch → dev	the engineer	on demand
merge to main → tst	CI service principal	automatic
tst → prd	prd service principal	gated manual action
anything → the UI	nobody	flagged as a defect

THE CI GATES

§9.5 · automated, not meetings

LEVEL	WHAT IS TESTED	THE GATE
Unit	transformation functions vs fixtures	must pass to merge
Contract	bronze schema assertions	source change fails the build
Quality	uniqueness of business key	quarantine or drop by policy
	referential integrity to dimensions	violation rates published
	domain-value validity, volume anomaly	
Reconcile	migrated measures vs the legacy EDW	zero unexplained variance, 30 d
Freshness	every certified product, monitored	a breach pages the owning pod
Performance	critical dashboard and query times	regression blocks release
Security	grants vs the declared model	hand-made grant is flagged

The one rule that carries the most weight

A table may not be tagged **certified** unless it has **expectations, a freshness monitor, an owner and a published contract**. Certification is the platform team's only real lever, and it must be used.

WHAT THIS BUYS — measured, not asserted (§12.2)

Deployment lead time	6–8 weeks → under 2 days merge → production
Pipeline first-run success	~89% est. → 99.5%
Changes authored by client engineers	0% → over 80% for 3 months — the handover gate

WORKSPACES, REGIONS AND METASTORES

§5.7

D-08 · THREE REGIONAL PRODUCTION WORKSPACES, ONE METASTORE PER REGION

EU metastore — regional managed storage
`mrg-eu-prd — EU/UK production (primary)`
`mrg-eu-nonprd — dev + test, bound to mrg_dev / mrg_tst`
`mrg-eu-sandbox — exploration, separate budget`

APAC metastore — regional managed storage
`mrg-apac-prd — production where residency requires in-region processing`

AMER metastore — regional managed storage
`mrg-amer-prd — Americas production`

A metastore is regional, so multi-region necessarily means multiple metastores. **Cross-region access is provided by OpenSharing, not by direct grants.** Confirm current cross-metastore options at design time **[TBC against current docs]**.

D-08 · the rejected alternatives

A single global workspace and metastore: simplest to operate, fails the residency test — several regulators care about processing location, not only storage location. **A workspace per country:** 14 workspaces, 14 sets of policies, and an operations team that spends its life on drift.

The boundary that actually matters

The workspace is a **working environment, not a security boundary** — Unity Catalog is the security boundary. Workspaces are allocated for **operational and residency reasons only** (§5.7).

Solution design · September 2026

Three environments identical in shape; the environment boundary is enforced by the catalog hierarchy, not by discipline (D-05, D-10).

Figure 4 of 6

Figure 4 — Environment and promotion

`case01_fig4_environments_and_promotion.svg`

FIGURE 5 Meridian Retail Group

Illustrative — fictional client

Security zones and the personal-data flow

where personal data enters, where the identifier is separated from the behaviour, who may re-join them, what crosses a border, and how an erasure request is executed and evidenced (§7.1–7.4, §7.6 · D-13, D-14)

WHERE PERSONAL DATA IS — and where, by construction, it is not

D-13 · pseudonymisation at the bronze-to-silver boundary, not at the consumption layer

SOURCES · by classification

P3 special category or payment data
POS tender detail / truncated PAN
HR absence reason, union membership

P2 personal data (GDPR Art. 4)
POS loyalty card number
E-commerce clickstream (device ID, IP, hashed email, customer ID)
Salesforce named contacts
Marketing platforms + consent store
HR Workday worker records
SAP vendor and employee-linked postings
Legacy EDW (P1/P2, poorly mapped)

P1 internal, commercially sensitive
Supply-chain EDI supplier terms
Third-party panel and price data

P0 no personal data
Product / PIM
IoT — DC and cold chain
Reference and finance master

CONTROLS AT INGEST

Tokenise or drop tender detail
P3 must **never land in bronze in the clear**. PCI scope is actively kept **out** of the lakehouse.

Join consent before any use
Consent arrives on a **separate topic**; bronze carries the join key and silver is filtered by consent at the point of creation.

Land in the resident region
EU data lands in EU storage under the EU metastore; APAC and Americas likewise.

BRONZE · the restricted zone

identifiers present

- Bronze retains the **source payload including direct identifiers** — this is deliberate.
- Readable by the **platform team and the ingestion identities only**. It is the most tightly held privilege on the platform.
- It is what makes the pipeline **replayable**, and therefore what makes a cautious single extract from SAP safe.
- Retention 13–25 months; **cold-chain bronze extended** to the food-safety record-keeping period **[TBC with Legal]**.

PSEUDONYMISATION BOUNDARY
D-13 · bronze → silver

SILVER · the identifier is separated from the behaviour

silver_customer_pii — RESTRICTED

Name, email, phone, address, loyalty card number, device ID — **extracted at the bronze-to-silver boundary into this one schema**. Readable by **two service principals and one small approval group**. Nothing else reads it.

silver_* · gold_* — pseudonymous

Everything downstream carries a **stable pseudonymous customer key with no derivable relationship to the identifier**. Analytics, features, models and shares use the surrogate. Most of the estate therefore simply **does not contain personal data** — a structurally stronger position than a masking policy that can be misconfigured.

GOLD · only under policy

Tagged, then masked by default
Where personal data must appear, the column is tagged **pii_type** and an **ABAC policy masks it by default**.

Unmasking is a group, with an expiry
Membership of **mrg-pii-approved-<domain>** with **a recorded justification and an expiry date**. Never in the default entitlement.

Re-identification happens in one place
A **governed join, executed by an approved process, logged every time**. Marketing's audience activation is such a process; an analyst's curiosity is not.

WHAT LEAVES · §7.6

~1,200 suppliers
`OpenSharing from gold_supplier_portal`
one recipient per supplier · row filter on the recipient's supplier identity · **share created without history** · no personal data · no competitor data

Media and creative agencies
`OpenSharing of audiences and performance`
pseudonymous keys and aggregates only · activation happens **through the marketing platform, not the share**

2 joint-venture partners
`OpenSharing, Databricks-to-Databricks`
agreed trading aggregates · contractually scoped · reviewed quarterly

External auditors
`Time-boxed read on gold_finance_reporting`
as-reported figures with provenance · expiring group · every access logged

Panel vendor (inbound)
`OpenSharing inbound where supported`
the licence's **redistribution restriction is an explicit exclusion from every outbound share, and it is tested**

RIGHT TO ERASURE — a pipeline, not a ticket

§7.3 · nine steps, in order

1 Request capture
`control table in ops_platform`
Requests land from the client's privacy request tool with subject identifier, timestamp and jurisdiction. A control table is required because requests arrive continuously and must be executed in batches.

2 Subject resolution
`resolve via silver_customer_pii`
Resolve the request to the **pseudonymous key(s)**. One person may hold multiple identifiers across countries — **this is the step most implementations get wrong**.

3 Scope determination
`lineage + data_classification = p2`
Enumerate every table holding the subject's data. **Lineage is the evidence that the scope is complete** — a governance capability, not a convenience.

4 Delete at the source layer first
`bronze, then propagate`
Databricks guidance is explicit that complete deletion is preferable to obfuscation, and that deleting at the source layer first is the correct order. **MERGE … WHEN MATCHED THEN DELETE** for many subjects in one run.

5 Propagate
`the detail everyone misses`
Materialised views handle source deletions automatically. **Streaming tables are append-only** — deletion needs DML plus a downstream read configured to skip change commits, or a full refresh (which risks loss if source retention expired). **Every streaming table holding personal data needs a documented erasure route agreed before production.**

6 Physically remove
`REORG TABLE … APPLY (PURGE) then VACUUM`
Deletion vectors mark rows deleted **without rewriting files**. Until **VACUUM** runs the data is still physically present — the erasure SLA must account for the retention threshold.

7 Close the copies
`time travel, clones and shares`
Shares default to including history on current runtimes, letting a recipient time-travel to a pre-deletion version. Shares carrying personal data must be created **without history**, and that must be a checked control, not a habit.

8 Evidence
`written back to ops_platform`
Request, resolution, tables affected, execution timestamps and completion, retained for the statutory period. **The evidence is the deliverable** — the deletion is assumed; the proof is what is audited.

9 Upstream
`Kafka, source files, source systems`
The obligation covers them too. Kafka retention (**7 days**, the §3.4 ask) bounds the exposure; source-system erasure remains the source owner's obligation and is tracked jointly.

COMMITTED SLA (illustrative)
erasure executed within 14 days · physical removal complete within 21 days · regulatory limit: one month
the gap between 14 and 21 days is the retention threshold — explain it to the DPO up front, rather than have it discovered during an audit

§7.1 · THE OBLIGATION LANDSCAPE — and what each one does to the architecture

OBLIGATION	WHERE IT BITES	ARCHITECTURAL CONSEQUENCE
GDPR, 14 countries + UK	loyalty, clickstream, marketing, HR, CRM	pseudonymisation, purpose limitation, erasure, RoPA
ePrivacy / consent	clickstream, marketing, app	consent state is a join condition, not a late filter
Residency and transfer	EU, UK, 3+ APAC jurisdictions	regional metastores and regional processing (D-08)
Works councils	5+ of the 14 countries	employee-level analysis gated on consultation
Statutory audit / controls	finance domain	immutable as-reported history, evidenced lineage, SoD
Food-safety record-keeping	cold chain	extended bronze retention, tamper-evident evidence trail
Payment card rules	POS tender data	keep it OUT of scope — tokenise or drop at the boundary
Third-party data licences	panel and competitor data	redistribution limits as grants and share exclusions

RESIDENCY AND AUDIT EVIDENCE

§7.4, §7.7 · D-08

§7.4 · RESIDENCY — five controls

CONTROL	IMPLEMENTATION
Storage	managed storage per regional metastore — EU data in EU storage
Processing	regional workspaces; jobs on EU-resident data run in the EU workspace
Cross-border	a global view is built from aggregated or pseudonymised regional
	products shared cross-region — never from raw personal data
Enforcement	the data_residency tag drives an ABAC row filter keyed on the
	analyst's regional group — a cross-region query returns only
	permitted rows rather than failing opaquely
Evidence	audit system tables record who read what, from where

⚠ An honest caveat
Residency has a real cost: some pipelines are built three times and some global questions are answered on aggregates rather than detail. Confirm with Legal, **per country**, which flows genuinely require in-region *processing* rather than in-region *storage* — the answer is usually narrower than the initial assumption, and each narrowing saves material effort.

§7.7 · WHAT AN AUDITOR ASKS, AND WHERE THE ANSWER COMES FROM

THE QUESTION	THE EVIDENCE
Who accessed personal data, and when?	audit system table on the restricted schema
What feeds this reported figure?	column-level lineage, captured automatically
Was this control in place on 3 March?	policy definitions in Git + deployment history
Has this erasure request completed?	erasure control table + the executing job's audit trail
Who changed this permission?	audit table, Unity Catalog service events
Was the reported number altered after?	append-only finance silver with restatement flags

⚠ Two limits to design around
Lineage covers **objects registered in Unity Catalog** — path-based access and renamed objects break it, which is another reason external paths are registered as volumes and external locations. Audit records are retained for a defined window by default: if the statutory obligation is longer, a **scheduled export of the audit tables into ops_platform is a Phase 1 deliverable**, not an afterthought. **[TBC]**

Solution design · September 2026

D-13: separating identifiers at the boundary means most of the estate simply does not contain personal data — a cheaper conversation with the DPO, and a much more targeted erasure.

Figure 5 of 6

Figure 5 — Security zones and data flow for personal data

`case01_fig5_security_zones.svg`

FIGURE 6 Meridian Retail Group

Illustrative — fictional client

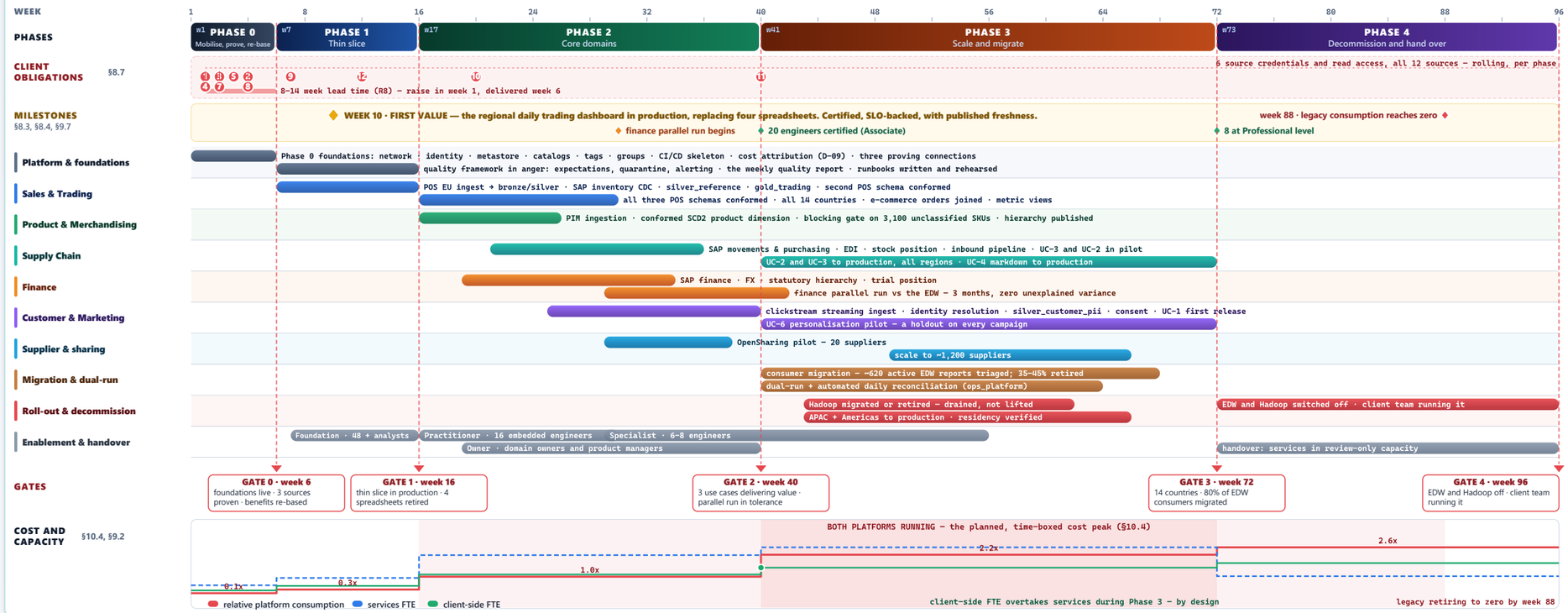
Delivery timeline: four phases, five gates, 96 weeks

first value at week 10 · the five core workstreams as swimlanes · the dual-run window and its planned cost peak · client obligations as dependencies with lead times (§8.1–8.7, §9.2, §10.4, §10.6)

week 1 → week 96

FOUR PHASES, FIVE GATES, 96 WEEKS

§8.1–8.7 · every phase ends in a decision point with defined evidence



CLIENT OBLIGATIONS — the real dependencies

§8.7 · numbered as pinned on the timeline

#	CLIENT OBLIGATION	NEEDED BY	CONSEQUENCE OF DELAY
1	Executive sponsor, cross-functional authority	week 1	every decision escalates
2	Named business owners for the 8 use cases	week 4	value unmeasurable
3	SAP basis availability + extraction decision	week 2	**critical path: 4 wk = 4 wk**
4	Network, private connectivity, firewall, DNS	w1 raise / w6	**everything is blocked**
5	Identity federation and group provisioning	week 3	workarounds become permanent
6	Source credentials, read access, 12 sources	rolling	per-source slip
7	DPIA and DPO engagement	week 2	Customer domain blocked
8	Works-council consultation initiated	week 4	HR domain deferred (assumed)
9	Client engineers embedded in pods	week 7	knowledge transfer fails
10	Legacy ETL documentation + its authors	week 20	archaeology cost doubles
11	Decision authority to retire reports	week 40	migration scope grows 40%
12	Kafka retention to 7 days + schema registry	week 12	clickstream reliability capped

Each one has a lead time that starts when it is asked for

— not when it is needed. **Network and connectivity (item 4) takes 8–14 weeks in a large enterprise** and is the most common cause of a late Phase 0 (R8). Design a temporary, security-approved path for Phase 0 proving work so that engineering is not idle.

DECOMMISSION GATES G1–G5

§8.6 · binding, per §1.5 ask 5

- **G1 No active consumers** — 90 consecutive days with zero queries from anything other than the reconciliation job
- **G2 Reconciliation clean** — every migrated measure at zero unexplained variance for 30+ days
- **G3 History preserved** — the retained subset exported, validated by row and checksum, and readable
- **G4 Regulatory retention satisfied** — Legal has confirmed in writing what must be kept, for how long, and where it now lives
- **G5 Rollback expired** — 60 days since the last cutover with no rollback invoked

All five, or it stays on

The legacy warehouse is switched off when, **and only when, all five are met**. The same five gates apply to the Hadoop cluster, **assessed separately**. Dual-run cost is a named, budgeted line item — the failure mode is not the cost, it is a **dual-run with no end date**, which is why G1–G5 are dated and owned.

THE STRANGLER, IN SEVEN STEPS (§8.6)

1. Identify the consumer and its **true owner**
2. Decide whether it should exist at all — **roughly 40% should not**
3. Trace its logic in the legacy ETL. This is archaeology: **3–5 days per non-trivial consumer, not 1**
4. Rebuild against the gold product — **a re-derivation, reviewed by the business owner**, not a lift of the SQL
5. **Dual-run**: both systems produce the number daily; reconciliation records and explains every variance
6. Cut over the consumer. **The legacy job keeps running**
7. When all consumers have cut over, disable the legacy job — **not delete** — for 30 days, then remove

VALUE REALISATION AND THE COST PEAK

§10.4, §10.6

§10.6 · PAYBACK — deliberately conservative

Uses the **lower end** of the benefit range, delays realisation by **two quarters** against the delivery plan, and assumes only **60% of the identified benefit is captured**.



Cumulative payback falls in the first half of Year 2

… even on this basis, and it remains inside Year 2 down to roughly **40% benefit capture**. The sensitivity that matters is **not cost** — cost is reasonably predictable — it is **whether the business changes its decisions**. A perfect forecast that does not change a safety-stock parameter is worth nothing.

PROGRAMME SHAPE (§10.4, §9.2)

PHASE	WKS	SERVICES	CONSUMPTION	LEGACY
0	6	~51 pw	0.1x	–
1	10	~150 pw	0.3x	–
2	24	~840 pw	1.0x baseline	legacy at 100%
3	32	~1,340 pw	2.2x	100%, reducing from w60
4	24	~396 pw	2.6x	retiring to zero by w88
Run	–	~4 FTE	~2.8x, flattening	zero

Solution design · September 2026

A programme without real gates is a programme that cannot be stopped — and one that cannot be stopped cannot be trusted (§8.1).

Figure 6 of 6

Figure 6 — Delivery timeline

`case01_fig6_delivery_timeline.svg`

6. Data products and domain ownership

6.1 How the estate is carved

Meridian is not ready for a full data mesh and should not attempt one. There is no existing culture of domain data ownership, and pushing accountability onto teams that have never held it produces orphaned datasets with a domain label on them. The recommended model is **federated ownership with a strong platform core**:

- A **central platform team** owns ingestion to bronze, Unity Catalog structure, security policy, CI/CD, cost management and the shared reference domain. Bronze is *plumbing* and should be built once, by specialists.
- **Domain teams** own silver and gold within their domain: the business logic, the quality rules, the contract and the roadmap.
- Domain ownership is **earned, not assigned**. A domain graduates from platform-built to domain-owned when it has a named owner, a product manager and at least one engineer who has passed the enablement path (§9.6). Until then the platform team runs it on the domain's behalf, explicitly and with an end date.

> `D-11` **Federated ownership with a strong platform core, not a pure mesh and not a central monolith.**
>
> **Why:** the central-monolith model is the current state and it produced a 340-item backlog; the pure-mesh model presumes capability that does not exist yet. **Rejected:** full data mesh from day one (fails on capability), and a central team owning all transformation (fails on throughput, and re-creates the bottleneck we were hired to remove).

6.2 The domains

Domain	Owner (client role)	Products published	Primary consumers
Sales & Trading	Director of Commercial Analytics	`gold_trading`: conformed sales fact (basket line), trading day aggregate, store performance, channel view, sales metric view	Finance, Commercial, Store Ops, Board reporting
Customer	Head of Loyalty & CRM	`gold_customer_analytics`: customer profile, segment assignment, lifetime value, consent state, audience sets	Marketing, Digital, Service, Loyalty
Product & Merchandising	Head of Master Data	`gold_merchandising`: conformed product dimension (SCD2), hierarchy, supplier linkage, price and markdown history	Every other domain — this is the most-depended-on product in the estate
Supply Chain	Director of Supply Chain Planning	`gold_supply_chain`: stock position by SKU-location, movement fact, inbound pipeline, forecast output, availability measure	Supply Chain, Store Ops, Commercial, E-commerce
Finance	Group Financial Controller	`gold_finance_reporting`: trial position, margin bridge, FX-converted reporting views, statutory hierarchy	Group Finance, regional controllers, Audit
Supplier	Head of Procurement	`gold_supplier_portal`: OTIF and quality scorecard, delivery detail, claims position — **shared externally**	Procurement, and ~1,200 suppliers via OpenSharing
Marketing	Marketing Operations Lead	`gold_customer_analytics` (shared with Customer): campaign performance, spend, attribution, suppression lists	Marketing, Agencies, Finance
People	HR Analytics Lead	`silver_people` only in Phases 1–3; gold deferred	Restricted; HR and Finance cost allocation
Platform (shared)	Head of Data Platform (services-led initially)	`silver_reference` (store, calendar, FX, hierarchy), `ops_platform` (quality results, cost attribution, lineage extracts, run metadata)	Everyone

6.3 What a data product must publish

A schema is not a data product. To be listed as a product in the catalogue, a schema must publish all seven of the following. The platform team refuses to certify anything missing an item, and that refusal is the mechanism by which quality is maintained.

#	Element	Form
1	**Owner**	A named group (`mrg-<domain>-owner`) and a named human product manager
2	**Purpose and grain**	One paragraph in the schema comment; one sentence per table stating its grain
3	**Schema contract**	Column names, types, nullability, and a documented change policy: additive changes are announced, breaking changes require a new version and a deprecation window (minimum 90 days)
4	**Quality guarantees**	The specific expectations enforced, the action on failure (drop, quarantine, fail), and the published pass rate
5	**Freshness SLO**	"Complete for day D-1 by 06:00 local, 99% of days", measured and published, not asserted
6	**Classification and residency**	Governed tags on every table and every personal-data column
7	**Tier**	`certified` (SLO-backed, supported, safe for executive reporting), `curated` (maintained, no SLO), `experimental` (no guarantees, no external consumption, auto-expiring)

Tier is a `product_tier` tag and it is visible everywhere the table is: in the catalogue, in the natural-language interface's scope, and in dashboards. A consumer should never have to ask whether a number can be trusted — the platform should already have told them.

6.4 How ownership maps to the catalog

Concept	Unity Catalog expression
Domain	A set of schemas (`silver_<domain>` , `gold_<domain>`) within each environment catalog
Domain ownership	`ALTER SCHEMA … OWNER TO mrg-<domain>-owner` — a group, always
Product contract	Schema and table comments, column comments, governed tags; machine-readable form generated into `ops_platform` for the catalogue
Publication	`GRANT USE SCHEMA, SELECT ON SCHEMA gold_<domain> TO mrg-<consumer>-reader`
Internal work	Silver schemas are granted only to the owning domain and to named downstream domain engineers, with each cross-domain grant recorded as a dependency
External publication	A share from `gold_supplier_portal` only; no other schema is shareable, enforced by policy and checked weekly
Deprecation	A table tagged `lifecycle = deprecated` with an end date in the comment; the weekly report names its remaining consumers from lineage

Cross-domain dependency rule. A domain may read another domain's **gold** products freely. Reading another domain's **silver** requires an explicit, recorded agreement, because it couples the two teams' release cycles. Reading another domain's **bronze** is prohibited. This one rule prevents the slow collapse of a layered architecture back into point-to-point coupling — which is precisely what §2.3 describes happening to the current estate.

7. Governance, security and compliance

7.1 The obligation landscape

Obligation	Where it bites	Architectural consequence
GDPR across 14 countries (plus UK GDPR)	Loyalty, clickstream, marketing, HR, Salesforce	Pseudonymisation, purpose limitation, erasure, records of processing
ePrivacy / consent	Clickstream, marketing, app	Consent state is a join condition, not a filter applied at the end
Data residency and transfer	EU, UK, and at least 3 APAC jurisdictions with local requirements	Regional metastores and regional processing (D-08); transfer mechanism documented per flow
Works councils	5+ countries	Employee-level analysis is gated on consultation (§3.10)
Statutory audit / financial controls	Finance domain	Immutable as-reported history, evidenced lineage, segregation of duties
Food safety record-keeping	Cold chain	Extended retention, tamper-evident evidence trail
Payment card rules	POS tender data	Keep it **out of scope** — tokenise or drop at the boundary
Third-party data licences	Panel and competitor data	Redistribution restrictions expressed as grants and share exclusions

7.2 PII handling and pseudonymisation

The design principle is **separate the identifier from the behaviour**.

1. Direct identifiers (name, email, phone, address, loyalty card number, device ID) are extracted at the bronze-to-silver boundary into a single restricted schema, `silver_customer_pii`, which is readable by two service principals and one small approval group. Nothing else reads it.
2. Everything downstream carries a **pseudonymous customer key** — a stable surrogate with no derivable relationship to the identifier. Analytics, features, models and shares use the surrogate.
3. Re-identification happens in exactly one place: a governed join, executed by an approved process, logged every time. Marketing's audience activation is such a process; an analyst's curiosity is not.
4. Where personal data must appear in an analytical table, the column is tagged and an ABAC policy masks it by default. Access to the unmasked value requires membership of a `mrg-pii-approved-<domain>` group with an expiry date and a recorded justification.
5. Bronze retains the source payload including identifiers, and is accessible only to the platform team and ingestion identities. This is deliberate: it is what makes the pipeline replayable, and it is why bronze access is the most tightly held privilege on the platform.

> `D-13` **Pseudonymisation at the bronze-to-silver boundary, not at the consumption layer. Why:** masking at consumption means the personal data is present in every downstream table, and the control is a policy that can be misconfigured. Separating identifiers at the boundary means most of the estate simply does not contain personal data — a structurally stronger position, and a much cheaper conversation with the DPO and with auditors. **Rejected:** full-estate masking policies over tables that contain identifiers everywhere (simpler to build, weaker to defend, and it makes erasure a search across the whole estate instead of a targeted operation).

7.3 Right to erasure — the mechanics

Erasure is a pipeline, not a ticket. The architect should treat the following as a design specification.

Step	Mechanism	Note
1. Request capture	Erasure requests land in a control table in `ops_platform` from the client's privacy request tool, with subject identifier, request timestamp and jurisdiction	A control table is required because requests arrive continuously and must be executed in batches
2. Subject resolution	Resolve the request to the pseudonymous key(s) via `silver_customer_pii`	One person may hold multiple identifiers across countries; this is the step that most implementations get wrong
3. Scope determination	Lineage from `silver_customer_pii` and the `data_classification = p2` tag enumerate every table holding the subject's data	Lineage is the evidence that the scope is complete; this is a governance capability, not a convenience
4. Delete at the source layer first	Delete in **bronze**, then propagate downstream — Databricks guidance is explicit that complete deletion is preferable to obfuscation, and that deleting at the source layer first is the correct order	A `MERGE … WHEN MATCHED THEN DELETE` is the recommended form when deleting many subjects in one run
5. Propagate	Materialised views handle source deletions automatically. **Streaming tables are append-only** — deletion requires either deleting from the streaming table with DML and configuring the downstream read to skip change commits, or a full refresh (which reprocesses everything and risks loss if source retention has expired). **Design for this now**: any streaming table holding personal data must have a documented erasure route agreed before it goes to production	This is the single most commonly missed detail in lakehouse GDPR designs
6. Physically remove	Deletion vectors mark rows deleted without rewriting files, so a `REORG TABLE … APPLY (PURGE)` is required to rewrite the affected files, followed by `VACUUM` to delete them	Until `VACUUM` runs, the data is still physically present. The erasure SLA must account for the retention threshold

Step	Mechanism	Note
7. Close the copies	Time travel (bounded by the log and deleted-file retention properties), clones, and **shares — which default to including history on current runtimes**, letting a recipient time-travel to a pre-deletion version	Shares carrying personal data must be created without history, and that must be a checked control, not a habit
8. Evidence	Record request, resolution, tables affected, execution timestamps and completion in `ops_platform`; retain for the statutory period	The evidence is the deliverable. The deletion is assumed; the proof is what is audited
9. Upstream	The obligation covers Kafka, source files and source systems too, not only the lakehouse	Kafka retention (7 days, §3.4) bounds the exposure; source-system erasure remains the source owner's obligation and must be tracked jointly

Committed SLA (illustrative): erasure executed within **14 days** of a verified request, with physical removal complete within **21 days**, against a regulatory limit of one month. The gap between 14 and 21 days is the retention threshold, and it must be explained to the DPO up front rather than discovered during an audit.

7.4 Residency

Control	Implementation
Storage residency	Managed storage per regional metastore; EU data in EU storage, and so on
Processing residency	Regional workspaces; jobs on EU-resident data run in the EU workspace
Cross-border analysis	Where a global view is required, it is built from **aggregated or pseudonymised** regional products shared cross-region, not from raw personal data
Enforcement	`data_residency` tag drives an ABAC row filter keyed on the analyst's regional group, so a cross-region query returns only permitted rows rather than failing opaquely
Evidence	Audit system tables record who read what, from where

An honest caveat: residency has a real cost. It means some pipelines are built three times and some global questions are answered on aggregates rather than detail. Confirm with Legal, per country, which flows genuinely require in-region processing rather than in-region storage — the answer is usually narrower than the initial assumption, and each narrowing saves material effort.

7.5 Access model for 400+ analysts

The model must survive the arrival of the 401st analyst without anyone doing anything.

Element	Design
Identity	Federated from the corporate directory; account-level groups synchronised; no local accounts
Provisioning	Directory group membership is the only entitlement mechanism. Joining `mrg-analyst-emea` grants a working set of access on the next sync, with no ticket
Default entitlement	Every analyst gets read on `certified` gold products in their region, plus the metric views. That default covers an estimated 80% of demand
Escalation	Additional access is a group join with an approver and an expiry, requested in the client's existing access tool
Personal data	Never in the default. Always a separate, expiring, justified group
Compute	Analysts are granted use of a warehouse class, never the ability to create compute
Review	Quarterly recertification driven from the audit tables — groups with no queries in 90 days are flagged for removal, and a group nobody uses is a group that should not exist
Onboarding	A new analyst is productive on day one via the default entitlement and the catalogue. If onboarding requires a conversation with the platform team, the model has failed

7.6 Sharing with suppliers and agencies

Recipient	Mechanism	Contents	Controls
~1,200 suppliers	**OpenSharing** from `gold_supplier_portal`	Each supplier sees only its own OTIF, quality and claims rows	One recipient per supplier; a row filter keyed on the recipient's supplier identity; **share created without history**; no personal data; no competitor data
Media and creative agencies	**OpenSharing** of audience definitions and campaign performance	Pseudonymous keys and aggregate performance only	No direct identifiers cross the boundary; activation happens through the marketing platform, not through the share
Joint-venture partners (2 markets)	**OpenSharing**, Databricks-to-Databricks where the partner is on the platform	Agreed trading aggregates	Contractually scoped; reviewed quarterly
External auditors	Time-boxed read on `gold_finance_reporting` plus lineage and audit extracts	As-reported figures with provenance	Expiring group; every access logged
Third-party panel data received	**OpenSharing** inbound where the vendor supports it	—	The licence's redistribution restriction is expressed as an explicit exclusion from every outbound share, and tested

D-14 **Supplier data exchange via OpenSharing, not a portal or per-supplier extracts. Why:** 1,200 suppliers times a file extract is an integration estate in its own right, and the current alternative (emailed spreadsheets) is both a governance failure and a source of disputes. Sharing is zero-copy, governed by the same access model, and within a region incurs no egress cost. **Rejected:** building a supplier web portal (a product with a roadmap, a support burden and an authentication estate, for a data-delivery problem), and per-supplier SFTP extracts (1,200 jobs and 1,200 ways to be wrong). Note that smaller suppliers will not consume a share directly; a thin UI over the same governed data serves them, reading the identical tables. Confirm cross-region and cross-cloud transfer cost implications for non-EU suppliers at design time **[TBC against current docs and cloud pricing]**.

7.7 Audit and lineage evidence

Question an auditor will ask	Where the answer comes from
Who accessed personal data, and when?	The audit system table, filtered by event date, on the restricted schema
What feeds this reported figure?	Table and column lineage, automatically captured, down to column level, covering notebooks, jobs, pipelines, dashboards and queries
Was this control in place on 3 March?	Policy definitions held in version control, with the deployment history as evidence
Has this erasure request been completed?	The erasure control table plus the audit trail of the executing job
Who changed this permission?	Audit table, Unity Catalog service events
Can you prove the reported number was not altered afterwards?	Append-only finance silver with restatement flags, plus table history within the retention window

Two practical limits the architect must design around: lineage covers objects registered in Unity Catalog — path-based access and renamed objects break it, which is another reason external paths are registered as volumes and external locations rather than accessed directly. And audit records are retained for a defined window by default; if the client's statutory obligation is longer, a scheduled export of the audit tables into `ops_platform` under the required retention is a **Phase 1 deliverable**, not an afterthought. Confirm current retention and any configurable options **[TBC against current docs]**.

8. Delivery plan

8.1 Shape of the programme

Four phases over 96 weeks. Each phase ends in a **gate**: a decision point with defined evidence, at which the programme can be stopped, re-scoped or continued. A programme without real gates is a programme that cannot be stopped, and one that cannot be stopped cannot be trusted.

Phase	Weeks	Theme	Gate evidence
0	1–6	Mobilise, prove, re-base	Platform foundations live; three source connections proven end to end; benefits re-based on client actuals; Phase 1 plan confirmed
1	7–16	Thin slice	One region, one domain, in production, with a real consumer using it in anger; decommission of at least 4 spreadsheets
2	17–40	Core domains	Customer, Product, Sales, Supply Chain and Finance silver and gold live for all regions; forecasting in pilot; 3 use cases delivering measured value
3	41–72	Scale and migrate	All 14 countries; 80% of active EDW consumers migrated; dual-run reconciliation passing; Hadoop workloads migrated or retired
4	73–96	Decommission and hand over	EDW and Hadoop off; client team running the platform; services role reduced to advisory

8.2 Phase 0 — Mobilise and prove (weeks 1–6)

Week	Deliverable
1	Cloud accounts, network design signed off, **private connectivity requests raised** (the long pole — start on day one). Identity federation design agreed
1–2	Landing-zone triage: what in the Hadoop cluster is worth keeping. Expect to recommend leaving 60%+ behind
2	Unity Catalog metastore, catalogs, initial schemas, group taxonomy, governed tags defined
2–3	CI/CD skeleton: repository structure, bundle definitions, dev/test/prod targets, first automated deployment
3–4	Three proving connections: SAP (one table group), POS EU (one country), PIM. Each landing into bronze, end to end
3–4	Cost attribution reporting live (D-09)
4–5	Benefits re-based: identity match-rate sample for UC-1, forecast-accuracy baseline for UC-2, availability measurement basis for UC-3
5	PII discovery pass over the in-scope EDW subset and the three proven sources
5–6	Phase 1 backlog agreed with named business owners; gate pack prepared

What "proven" means: data lands, is queryable, has a quality expectation attached, has lineage, is tagged, appears in cost reporting, and was deployed by the pipeline rather than by hand. Anything less is a demonstration, not a proof.

8.3 Phase 1 — The thin slice (weeks 7–16), first value at week 10

The thin slice is **one region (EU), one business outcome (daily trading and stock position), one consumer group (regional Commercial and Supply Chain)**. It is deliberately narrow and deliberately vertical: it exercises every layer of the architecture rather than building one layer broadly.

Week	Deliverable
7–8	POS EU ingestion at full country scope, bronze to silver conforming for one of three POS schemas
7–9	SAP inventory movements CDC into bronze and silver

Week	Deliverable
8–9	`silver_reference` (store master, calendar, FX) with a named owner and a maintenance process
9	`gold_trading` first product: daily sales by store, SKU and day, with margin
10	**First value: the regional daily trading dashboard in production**, replacing four spreadsheets. Certified, SLO-backed, with published freshness
10–12	Stock position product; availability measure defined and agreed (expect a fortnight of argument about the definition — budget for it, it is real work)
12–14	Quality framework in anger: expectations, quarantine, alerting, the weekly quality report
14–15	Second POS schema conformed (proves the pattern generalises)
15–16	Gate pack: measured freshness, measured quality, measured cost per product, user adoption, and a candid list of what was harder than expected

Why week 10 and not week 6. A dashboard can be produced in three weeks. A dashboard that is governed, tested, SLO-backed, cost-attributed, deployed from version control and owned by a named business team takes ten. The second one takes two. Phase 1 buys the *second* one.

8.4 Phase 2 — Core domains (weeks 17–40)

Five workstreams run in parallel, each with its own domain owner:

Workstream	Weeks	Lands
Sales & Trading	17–30	All three POS schemas conformed; all 14 countries; e-commerce orders joined; `gold_trading` complete with metric views
Product & Merchandising	17–26	PIM ingestion; conformed SCD2 product dimension; the "to be classified" quality gate; hierarchy published
Supply Chain	22–36	SAP movements and purchasing; EDI; stock position; inbound pipeline; UC-3 in pilot; UC-2 forecast in pilot
Finance	20–34	SAP finance; FX; statutory hierarchy; trial position; **parallel run against the EDW begins at week 30**
Customer	26–40	Clickstream streaming ingest; identity resolution; `silver_customer_pii` and pseudonymisation; consent integration; UC-1 first release

Cross-cutting in the same window: supplier scorecard and the first OpenSharing pilot with 20 suppliers (weeks 30–38); analyst onboarding at scale from week 24; the enablement programme (§9.6) running continuously from week 17.

Gate at week 40: three use cases delivering measured value; finance parallel run showing variance inside tolerance; at least two domains operating with genuine client-side ownership.

8.5 Phase 3 — Scale and migrate (weeks 41–72)

This is where the programme either succeeds or becomes permanent. Discipline is everything.

Workstream	Weeks	Lands
Consumer migration	41–68	The ~620 active EDW reports triaged: migrate, rebuild, merge or retire. Expect **35–45% to be retired outright** — that is the normal finding and it must be presented as a win
Dual-run and reconciliation	41–64	Automated daily reconciliation between EDW and lakehouse for every migrated measure
Hadoop migration	44–62	Workloads migrated or retired; retained history moved; the cluster drained rather than lifted
Use case scale-out	41–72	UC-2 and UC-3 to production across all regions; UC-4 markdown to production; UC-6 personalisation pilot
Regional roll-out	44–66	APAC and Americas workspaces to full production; residency controls verified per country
Supplier sharing	50–66	Scale from 20 to 1,200 suppliers

8.6 Migration approach — strangler, dual-run, decommission gates

The strangler pattern, applied concretely. The EDW is not migrated. Its *consumers* are migrated, one at a time, to products built from source. Each migration follows the same seven steps:

1. Identify the consumer (a report, an extract, a downstream system) and its true owner.
2. Determine whether it should exist at all. Roughly 40% should not.
3. Trace its business logic in the legacy ETL and document it. This is archaeology, and it is the expensive part — budget 3–5 days per non-trivial consumer, not 1.

4. Rebuild the logic against the lakehouse gold product. Not a lift of the SQL — a re-derivation, reviewed by the business owner.
5. **Dual-run** for a defined period: both systems produce the number daily, and an automated reconciliation job compares them, records the variance, and explains every difference above tolerance.
6. Cut over the consumer. The legacy job keeps running.
7. When all consumers of a legacy job have cut over, the legacy job is disabled — not deleted — for 30 days, then removed.

Reconciliation is a product, not an activity. Build it in Phase 2, before it is needed: an automated job, a reconciliation results table in `ops_platform`, a dashboard showing every migrated measure's variance over time, and a rule that a measure must show **zero unexplained variance for 30 consecutive days** before its consumer may cut over. "Explained" means written down, agreed and attributed to a specific cause — usually a defect in the legacy logic, which is itself a valuable finding to publish.

Decommission gates. The legacy warehouse is switched off when, and only when, all five are met:

Gate	Evidence
G1 — No active consumers	90 consecutive days with zero queries from anything other than the reconciliation job
G2 — Reconciliation clean	Every migrated measure at zero unexplained variance for 30+ days
G3 — History preserved	The retained history subset exported, validated by row and checksum, and readable
G4 — Regulatory retention satisfied	Legal has confirmed, in writing, what must be kept and for how long, and where it now lives
G5 — Rollback expired	60 days since the last cutover with no rollback invoked

The same five gates apply to the Hadoop cluster, assessed separately.

Dual-run cost is a named, budgeted line item, not an overrun. Both platforms run in parallel from the first migrated measure in **week 41** until legacy consumption reaches zero at **week 88** — a 47-week overlap, of which weeks 41–64 also carry the build and daily operation of automated reconciliation (§8.5). That is expensive and entirely predictable; §10.4 carries it explicitly. The failure mode here is not the cost — it is a dual-run with no end date, which is why G1–G5 are dated and owned.

8.7 What the client must provide

The programme cannot deliver these, and each one has a lead time that starts when it is asked for, not when it is needed.

#	Client obligation	Needed by	Consequence of delay
1	Executive sponsor with cross-functional authority	Week 1	Prioritisation stalls; every decision escalates
2	Named business owners for each of the 8 use cases	Week 4	Value is unmeasurable and unattributable
3	SAP basis team availability and an extraction decision	Week 2	Critical path; a 4-week delay here is a 4-week delay to Phase 2
4	Network and private connectivity, firewall and DNS	Week 1 request, week 6 delivery	Everything is blocked. This is the most common cause of a late Phase 0
5	Identity federation and group provisioning	Week 3	Access model cannot be implemented; workarounds become permanent
6	Source system credentials and read access, all 12 sources	Rolling, per phase	Per-source slip
7	Data protection impact assessment and DPO engagement	Week 2	Customer domain (Phase 2) blocked
8	Works-council consultation initiated	Week 4	HR domain deferred (already assumed)
9	Client-side engineers embedded in pods, from Phase 1	Week 7	Knowledge transfer fails; the programme never ends
10	Legacy ETL documentation and access to the people who wrote it	Week 20	Migration archaeology cost doubles
11	Decision authority to retire reports	Week 40	Migration scope grows by 40%
12	Kafka retention increase to 7 days and a schema-registry commitment	Week 12	Clickstream reliability capped; incidents recur

9. Team, ways of working and enablement

9.1 Pod shape

The delivery unit is a **pod**: 6–8 people, one domain, end-to-end accountability from source to consumer. Pods do not hand work between themselves. A pod that cannot deliver a product without another pod's sprint capacity is wrongly drawn.

Role	Per pod	Responsibility
Pod lead / senior data engineer	1	Technical accountability, design review, mentoring
Data engineer	2–3	Pipelines, quality, tests
Analytics engineer	1	Silver-to-gold modelling, metric definitions, dashboards
Client-side engineer (embedded)	1–2	Learns by building; owns the product after handover
Business analyst / product owner (client)	0.5–1	Requirements, acceptance, adoption

Plus a standing **platform pod** (not per-domain): 4–6 people owning Unity Catalog, CI/CD, security policy, cost, observability and the shared reference domain. The platform pod's customers are the other pods, and its success measure is how little the other pods have to think about the platform.

9.2 Rough FTE by phase (illustrative)

Role	Ph 0 (w1–6)	Ph 1 (w7–16)	Ph 2 (w17–40)	Ph 3 (w41–72)	Ph 4 (w73–96)
Engagement lead / principal architect	1.0	1.0	1.0	0.8	0.5
Solution architect	1.0	1.0	1.5	1.5	0.5
Platform engineers	2.0	3.0	4.0	4.0	2.0
Data engineers	2.0	5.0	14.0	16.0	6.0
Analytics engineers	0.5	2.0	5.0	6.0	2.0
ML engineers / data scientists	0	0	3.0	4.0	1.0
Data governance / security specialist	1.0	1.0	1.5	1.5	0.5
Migration specialists (EDW/Hadoop)	0.5	0.5	2.0	5.0	2.0
Delivery manager / scrum	0.5	1.0	2.0	2.0	1.0
Enablement lead	0	0.5	1.0	1.0	1.0
Services total	**8.5**	**15.0**	**35.0**	**41.8**	**16.5**
Client-side (required)	**4.0**	**8.0**	**18.0**	**24.0**	**28.0**

The crossover matters more than the totals. Client-side FTE overtakes services FTE during Phase 3 by design. **If that crossover does not happen, the engagement has failed regardless of what was built.**

9.3 Client-side roles to be filled

Role	FTE	Note
Head of Data Platform	1	Accountable owner from week 1. Hire or appoint before the programme starts, not during it
Domain data owners	8 x 0.3	Existing business leaders, part-time, named
Domain product managers	5	New or repurposed; the most commonly missing role
Data engineers (embedded, then owning)	12–16	Largely from the existing 48-person team, retrained
Platform / DevOps engineers	4	Own CI/CD, cost and infrastructure after handover
Data governance lead	1	Works with the DPO; owns the tag taxonomy and access model
Analytics engineers	6–8	Sit with the domains, not in a central pool

9.4 CI/CD and environments

`D-10` **All platform objects and workloads are deployed as Declarative Automation Bundles from Git.** Jobs, pipelines, schemas, volumes, dashboards, quality monitors and grants are defined in configuration in the repository and deployed by the pipeline. **Why:** the current estate's inability to answer "what changed" (§2.3) is the root cause of most of its incidents; code in version control with a deployment history answers it by construction, and it makes the three environments genuinely identical in shape. **Rejected:** creating jobs in the UI and promoting by export/import — faster on day one, and it guarantees environment drift, undocumented production changes, and a platform nobody can rebuild.

Element	Design
Repository	Monorepo, one directory per domain, shared libraries in `common/`; trunk-based with short-lived feature branches
Targets	`dev` (development mode: prefixed resources, paused schedules, per-developer isolation), `tst`, `prd` (production mode: service principal identity, Git branch validation, no cluster overrides)
Identity	Deployment to test and production runs as a service principal using workload identity federation from the CI system — **no long-lived tokens in CI**
Promotion	Merge to main deploys to test automatically; production deployment is a gated action from the same commit
Environment data	Dev holds a representative, masked subset — not 4% of prod with a different schema (§2.3). Producing that subset is a Phase 0 deliverable
Drift detection	A weekly job compares deployed state against the repository and against the declared grant model; drift is a defect with an owner

9.5 Testing and quality gates

Level	What	Gate
Unit	Transformation functions tested in isolation against fixtures	Must pass to merge
Contract	Bronze schema assertions; a source schema change fails the build rather than production	Must pass to merge
Data quality	Pipeline expectations on every silver table: uniqueness of the business key, referential integrity to conformed dimensions, domain-value validity, volume anomaly	Failing rows quarantined or dropped by declared policy; violation rates published
Reconciliation	Migrated measures compared to the legacy system daily	Zero unexplained variance for 30 days before cutover
Freshness and completeness	Automated monitoring on every certified product	Breach pages the owning pod
Performance	Critical dashboard and query response times tracked over time	Regression beyond threshold blocks release
Security	Grant model validated against the declared model; a hand-made grant is flagged	Weekly; exceptions require sign-off

The one rule that carries the most weight: a table may not be tagged `certified` unless it has expectations, a freshness monitor, an owner and a published contract. Certification is the platform team's only real lever, and it must be used.

9.6 Runbooks

Written during Phase 1, not after go-live, and rehearsed:

Runbook	Covers
Late or missing source file	Detection, tolerance, backfill, communication to affected consumers
Source schema change	Triage, rescued-data inspection, contract update, downstream impact from lineage
Streaming pipeline failure	Checkpoint inspection, restart, replay, gap verification
Quality expectation breach	Quarantine review, root cause, reprocess, consumer notification
Erasure request execution	The nine steps in §7.3
Access request and revocation	Group model, approval, expiry
Cost anomaly	Attribution query, top-consumers report, throttle and escalate
Peak trading readiness	Pre-peak checklist, capacity review, freeze window, on-call rota
Reconciliation variance	Investigation protocol, explanation register, escalation
Disaster recovery	RPO/RTO per zone, restore procedure, verification. Targets **[TBC with the client]**

9.7 Enablement and certification path

Capability transfer is a deliverable with an acceptance criterion, not a goodwill gesture.

Stage	Audience	Content	Timing
Foundation	All 48 data team members, plus analysts	Lakehouse concepts, Unity Catalog, Databricks SQL, the catalogue and how to find things	Weeks 8–16
Practitioner	16 embedded engineers	Declarative pipelines, Auto Loader and streaming ingest, testing, bundles and CI/CD, cost engineering	Weeks 17–30, paired with real delivery work
Specialist	6–8 engineers	Streaming at scale, performance and cost optimisation, security and ABAC, migration patterns	Weeks 30–56
Owner	Domain owners and product managers	Data product thinking, contracts, SLOs, quality, the certification process	Weeks 20–40
Certification	Target: 20 engineers certified (Associate level) by week 40; 8 at Professional level by week 72		

Enablement is delivered **in the pods, on the real backlog** — classroom training on a sample dataset produces people who can pass an exam and not a change. The reference poster series (47 verified visual references covering the platform, ingestion, Delta Lake, governance, cost and production engineering) is used as the standing desk reference and as the backbone of the certification track.

Acceptance criterion for handover: for three consecutive months, more than 80% of production changes are authored, reviewed and deployed by client engineers, with services in review-only capacity.

10. Cost model and commercials

> **No list prices appear in this section.** Databricks and cloud pricing change, vary by region, cloud, commit level and contract, and are the wrong thing to hard-code into a business case. What follows is the **structure** of the cost: the drivers, the ratios we typically observe, and the controls. A real estimate is produced by applying the client's measured workload profile to the current price lists during Phase 0 — and it is produced jointly with the Databricks account team and the cloud provider, because commit structure materially changes the answer.

10.1 The cost drivers

Driver	What moves it	Typical share of run cost
Platform consumption — data engineering (pipelines, jobs, streaming)	Volume ingested, transformation complexity, streaming vs batch, trigger frequency, whether pipelines are always-on	35–50%
Platform consumption — SQL and BI	Number of analysts, dashboard refresh frequency, query efficiency, warehouse auto-stop discipline, how much gold is pre-aggregated	25–40%
Platform consumption — ML (training, features, serving)	Model count, retraining frequency, serving request volume and latency target	5–15%
Cloud storage	Total retained volume, retention policy, file layout, how aggressively maintenance runs	5–10%
Cloud egress and networking	Cross-region replication, cross-cloud sharing, private connectivity	2–8%, and the most commonly forgotten
Ingestion-specific fixed costs	A CDC ingestion gateway runs on classic compute in the client's network and is **billed continuously, including while the ingestion pipeline is idle** — this makes the SAP connector a standing cost, not a per-run one	Material for SAP; budget it explicitly
Migration effort (one-off)	Number of consumers, archaeology depth, reconciliation scope	See §10.4
Dual-run (one-off, time-boxed)	Legacy platform run cost for the overlap period, plus the reconciliation workload	See §10.4

The three levers that move platform consumption most, in our experience, in order:

1. **Trigger frequency and always-on pipelines.** The gap between a continuously running stream and the same pipeline on a scheduled available-now trigger is very large, and for most workloads the business cannot tell the difference. This is D-04, and it is the single biggest lever.
2. **Bytes scanned.** Good file layout (clustering on the real filter columns), automatic table maintenance, and materialising the aggregates that dashboards actually query. A dashboard that scans a year of basket lines on every refresh is the classic and expensive mistake.
3. **Idle compute.** Auto-stop, serverless where demand is bursty, and never letting analysts create their own long-running compute.

10.2 How we estimate

Step	Method
1	Profile actual volumes per source (§3 gives the shape; Phase 0 measures the reality)
2	Classify each pipeline by mode and frequency from §5.4; count always-on pipelines separately, because they are priced by wall-clock, not work
3	Benchmark a representative pipeline per class in Phase 0, on real data, and measure consumption directly
4	Extrapolate by volume, with a peak factor (2.4x for POS at Christmas; 3x for clickstream at Black Friday) applied to the streaming and SQL classes
5	Model the analyst population's query behaviour from the legacy warehouse's query logs — not from a guess about "400 analysts"
6	Apply the current price lists per SKU, per cloud, per region, with the client's commit structure
7	Add cloud storage on the retained-volume projection and egress on the cross-region flows
8	Add a 25% contingency in Year 1, reducing to 10% in Year 3 as measurement replaces estimation

Steps 3 and 5 are what separate a defensible estimate from a guess. **Do not price this engagement without them**, and say so to the client.

10.3 How we control cost

Five controls, all from §5.7, restated here as commercial commitments:

Control	Commitment
Enforced tagging	100% of consumption attributable to a cost centre, domain and environment from week 4. Untagged consumption is treated as a defect
Budgets and alerts	Per-domain budgets with threshold alerts; reviewed weekly for six months, monthly thereafter
Attribution reporting	A cost-per-data-product report, published monthly to domain owners. Owners who see their bill manage it; owners who never see it never do
Efficiency review	A monthly review of the ten most expensive pipelines, queries and dashboards, with a named owner and an action for each
Retirement	Products with no consumption for 90 days are flagged; certified products with no consumers are decertified. **Retirement is a cost control, and nobody does it unless it is scheduled**

10.4 Illustrative effort and cost profile

Effort in person-weeks from §9.2. **Cost columns are illustrative placeholders** for the shape of the profile, not a quotation.

Phase	Weeks	Services person-weeks	Platform consumption (relative)	Legacy dual-run	Notes
0	6	~51	0.1x	—	Small footprint; benchmarking workloads only
1	10	~150	0.3x	—	One region, one domain
2	24	~840	1.0x (baseline)	Legacy continues at 100%	Five parallel workstreams; the peak of services effort
3	32	~1,340	2.2x	Legacy at 100%, reducing from week 60	**Peak total cost.** Both platforms running, migration at full rate
4	24	~396	2.6x	Legacy retiring to zero by week 88	Consumption rises as adoption rises; services fall sharply
Run (Yr 3+)	—	~4 FTE advisory	~2.8x, then flattening	Zero	Steady state

Two things to say plainly to the sponsor:

- **Phase 3 is the expensive one**, because both platforms run at once. This is visible, planned and time-boxed by the gates in §8.6. A programme that hides this is a programme that will have an uncomfortable conversation in month 14.
- **Platform consumption keeps rising after delivery stops**, because success means more people using more data. That is the right outcome, and it is why per-product cost attribution matters more than a total budget number. The metric to manage is **cost per unit of consumption**, not total cost.

10.5 The commercial construct we would propose

Phase	Construct	Rationale
0	Fixed price	Scope is well defined; the client should not carry discovery risk
1	Fixed price against a defined thin slice	We should carry the risk of our own first-delivery estimate
2	Capped time and materials per workstream, with per-product acceptance	Scope genuinely emerges; a fixed price here is priced with a risk premium the client pays regardless
3	Capped time and materials, plus a decommission-linked element	Aligns our incentive with the client's actual goal, which is switching the legacy platform off
4	Reducing retainer, stepped down quarterly against the handover criterion in §9.7	Makes our exit a contractual event rather than a conversation

10.6 Payback logic

Illustrative, and deliberately conservative — it uses the **lower** end of the benefit range, delays benefit realisation by two quarters against the delivery plan, and assumes only 60% of the identified benefit is actually captured.

	Year 1	Year 2	Year 3
Benefit realised (60% of low case, phased)	~$18m	~$79m	~$129m
Programme cost (services + platform + dual-run)	~$16m	~$18m	~$8m
Net	~+$2m	~+$61m	~+$121m
Cumulative	~$2m	~$63m	~$184m

Cumulative payback falls in the first half of Year 2 even on this conservative basis. The sensitivity that matters most is not cost — cost is reasonably predictable — it is **whether the business changes its decisions**. A perfect forecast that does not change a safety-stock parameter is worth nothing. That is why §12 measures business outcomes, not platform outputs, and why §8.7 asks for named business owners in week 4.

11. Risks and mitigations

Likelihood and impact on a 1–5 scale. These are the ten that actually kill programmes of this shape — not a generic register.

#	Risk	L	I	Mitigation	Owner
R1	**Source access is not granted in time.** Credentials, firewall rules and read permissions arrive weeks after they were promised, and Phase 0 becomes Phase 0 plus six	4	4	A source-access RACI agreed in week 1 with named owners and dates; access tracked as a standing agenda item; a blocked source is escalated to the sponsor within 5 working days, automatically, not by judgement	Client CIO / Engagement lead
R2	**SAP extraction politics.** The basis team refuses log-based CDC on the production ERP; the conversation runs for months	4	5	Engage basis in week 1, not week 8, and bring the cost of the alternative with you. Cost and design the ODP/SLT fallback in Phase 0 so a refusal is a plan change, not a re-architecture. Escalation path pre-agreed to the CFO (§1.5, ask 3)	Client CIO / Solution architect
R3	**Data quality is materially worse than believed.** Match rates, book-stock accuracy and hierarchy completeness turn out well below assumption, and use-case value collapses	4	4	Measure the three load-bearing assumptions in Phase 0 (identity match rate, forecast accuracy baseline, stock accuracy) on real samples. Re-base the benefit case at the week-6 gate and say so publicly. Quality remediation is scoped as its own workstream with a business owner, not absorbed silently	Solution architect / Domain owners
R4	**Dual-run cost triggers a budget crisis in month 12–16.** Both platforms running at peak, with benefits not yet fully landed	3	4	Budget dual-run explicitly from day one (§10.4) and show the peak on the first cost slide. Bind the decommission gates (§8.6) with dates and owners. Report the legacy platform's remaining consumer count monthly, so the end is visibly approaching	CFO / Programme director
R5	**Skills gap.** The client team cannot absorb the platform at the rate the plan assumes; handover slips and the services engagement becomes permanent	4	4	Embed client engineers in pods from week 7, not week 40. Enablement runs on the real backlog (§9.6). Make the 80%-client-authored-changes criterion a contractual handover gate. Accept that 10–15 of the 48 existing team members will not make the transition, and plan for that honestly rather than discovering it	Client Head of Data Platform / Enablement lead
R6	**Scope creep from the BI backlog.** The 340-item backlog arrives in Phase 2 disguised as requirements; the programme becomes a report factory	5	3	Backlog triage is an explicit Phase 3 workstream with a retirement target, not a Phase 2 intake. A new request must attach to one of the eight use cases or wait. Publish the retirement count as a headline metric so that *not* building something is visibly a success	Programme director / Domain product managers
R7	**PII discovery finds more than expected**, particularly in the legacy EDW and in clickstream, triggering a DPO intervention mid-flight	3	4	Automated classification scan in Phase 0 across the in-scope EDW subset and every source before it is ingested. DPO engaged from week 2 and attends the phase gates. Assume the answer is "more than we thought" and build the pseudonymisation boundary (§7.2) before Phase 2, not after	Data governance lead / DPO
R8	**Private connectivity and network lead times.** Circuits, endpoints, firewall change windows and DNS take 8–14 weeks in a large enterprise, and nobody counted them	4	5	Raise every network request in week 1 (§8.7, item 4). Design a temporary, security-approved path for Phase 0 proving work so that engineering is not idle. Track connectivity as a critical-path item on the programme plan with its own RAG	Client CIO / Platform lead
R9	**Vendor lock-in concerns stall the decision**, usually raised late and by someone not previously in the room	3	3	Address it in the architecture, not in a slide: open table format, data in the client's own cloud storage account, open sharing protocol for external exchange, transformation logic in SQL and Python in the client's own repository, infrastructure as code. State plainly what *is* proprietary and what exit would actually cost. A credible, costed exit path is more persuasive than a claim that lock-in does not exist	Principal architect
R10	**Change fatigue.** 2,400 stores, 85,000 employees, and a business already absorbing other change. Adoption stalls and the platform becomes a well-built thing nobody uses	4	4	Sequence the thin slice at a team that has actively asked for it. Retire the replaced spreadsheet on cutover day — do not leave both running, because both running means neither is trusted. Measure adoption weekly (§12) and treat a flat adoption curve as a delivery defect. Name a business change lead, funded, from Phase 1	Executive sponsor / Business change lead

Two risks worth watching that are not in the top ten: peak-trading readiness (the first Christmas on the new platform is a real event and needs a rehearsal, a freeze window and an on-call rota), and key-person dependency on the two or three people who understand the legacy ETL — interview and document them early, because they are the most likely to leave during the programme.

12. Success metrics

Two panels. The executive panel measures whether the business changed. The platform panel measures whether the platform is trustworthy. Both are published; the second is the leading indicator of the first.

12.1 Business KPIs

KPI	Baseline	Target (by)	How measured
On-shelf availability	94.1%	96.0% (end Yr 2), 96.8% (end Yr 3)	Existing availability audit process, unchanged, so the comparison is honest
Inventory value	$3.4bn	$3.05–3.15bn at constant sales	SAP inventory valuation, monthly average
Forecast accuracy (WMAPE, SKU-store-day)	~38%	26–30% (end Yr 2)	Held-out backtest on a fixed evaluation set, re-run monthly
Markdown as % of revenue	5.6%	5.2–5.4% at constant sell-through	Finance reporting; sell-through controlled for
Month-end close	9 working days	4 working days (end Yr 2)	Date of final signed trial position
Finance reconciliation effort	~14 FTE/month	4–5 FTE/month	Time recording, existing process
Customer identity match rate	Unknown	Measured in Phase 0, then 70–80%	Match rate between store loyalty and e-commerce identity within the loyalty population
Loyalty-attributed sales share	44%	48% (end Yr 3)	POS loyalty capture rate
Supplier OTIF	87.4%	92% (end Yr 3)	Scorecard, shared with suppliers, so the number is agreed rather than argued
Cold-chain excursion write-off	Baseline in Phase 2	-30%	Quality system, linked to sensor evidence
Active reports in use	~620 actively used, of ~2,900 registered	~400 certified data products; all 2,900 legacy reports retired	Catalogue and usage telemetry. **Fewer is the target** — the 2,900 are replaced by the 400, not added to
Time from question to answer (new analysis)	Median 11 months (backlog age)	Median under 5 working days for self-service; under 6 weeks for a new product	Request tracking

12.2 Platform SLOs

SLO	Baseline	Target	How measured
Certified product freshness	Not measured	Day D-1 complete by 06:00 local, 99% of days	Freshness monitoring on every certified product; breaches logged
Real-time inventory latency	24 h+	Under 15 min, 99th percentile	End-to-end timestamp from POS event to queryable stock position
Cold-chain alert latency	Not measured	Under 60 s, 99.9%	Sensor timestamp to alert dispatch
Pipeline success rate	~89% (est.)	99.5% first-run success	Job run telemetry
Data quality pass rate	Not measured	Over 99.5% of rows passing expectations on certified products; 100% of breaches investigated	Pipeline expectation metrics, published weekly
Incidents per month	~99	Under 15 by end Yr 2	Incident management
Firefighting share of engineering time	~70%	Under 25% by end Yr 2	Time allocation. **This is the metric the engineering team will care about most, and it should be published**
Mean time to restore (P1)	Not measured	Under 2 h	Incident management
Cost attribution coverage	0%	100% of consumption tagged	Billing telemetry; untagged consumption reported as a defect

SLO	Baseline	Target	How measured
Cost per certified product	Not measurable	Published monthly; trending down per unit of consumption	Attribution reporting
Analyst self-service rate	~15% (est.)	Over 70% of questions answered without engineering involvement	Request telemetry vs query telemetry
Time to onboard a new analyst	3–4 weeks	Under 1 day	Group membership to first successful query
Lineage coverage	0%	100% of certified products with end-to-end column lineage	Lineage system tables
Erasure SLA	Manual, unmeasured	100% executed within 14 days; physical removal within 21	Erasure control table
Deployment lead time	6–8 weeks	Under 2 days from merge to production	CI/CD telemetry
Changes authored by client engineers	0%	Over 80% for three consecutive months (handover gate)	Repository telemetry

13. What we would draw next

Six diagrams, in this order. Each is drawn from the sections named, so they stay consistent with the case rather than drifting into decoration.

#	Diagram	It must show	Drawn from
1	**End-to-end flow**	All 12 sources on the left, the three medallion zones, the serving layer, and the consumer groups on the right. Streaming paths visually distinct from batch. Each arrow labelled with its latency requirement, so the picture answers "how fresh is this" without a legend	§3.0, §5.2, §5.4, §5.6
2	**Per-source ingestion map**	One row per source: source, protocol, ingestion mechanism, landing object, volume and latency. This is the architect's build sheet and should be readable as a checklist. Colour-code by mechanism (managed connector / Auto Loader / streaming / federation / sharing / COPY INTO)	§3.1–3.12, §5.3
3	**Unity Catalog object model**	The three environment catalogs, the schema-per-domain layout, volumes and external locations, the group taxonomy, and which grants flow where. Overlay the governed tags and show which ABAC policy attaches at which level	§5.5, §6.4
4	**Environment and promotion**	Repository to dev to test to production, with the bundle deployment path, the identity used at each step, and the gates between. Show what is deployed (jobs, pipelines, schemas, grants, dashboards) and what is not	§9.4, §9.5
5	**Security zones and data flow for personal data**	The pseudonymisation boundary, the restricted PII schema, which zones contain personal data and which do not, the erasure path through all nine steps, and the residency boundaries between the three regions. This is the diagram Legal and the DPO will actually read	§7.2, §7.3, §7.4
6	**Delivery timeline**	The four phases on a week axis, the five workstreams as swim lanes, the gates as vertical markers, the thin-slice milestone at week 10, the dual-run window with its cost peak, and the five decommission gates. Client obligations (§8.7) shown as dependency markers with their lead times	§8.1–8.7, §9.2

Two further diagrams worth drawing if the audience warrants them: a **use-case dependency graph** (which use case needs which data product, and therefore what the true build order is — §4, §6.2), and a **cost-driver model** (where consumption arises and which control governs each driver — §10.1–10.3).

14. Appendix

14.1 Glossary — client-facing terms

Term	Plain meaning
Lakehouse	One place that holds all the data, from raw to reporting-ready, with one set of access rules. It replaces the separate "data lake for everything" and "warehouse for reporting" split
Medallion (bronze / silver / gold)	Three stages of refinement. Bronze is data exactly as it arrived. Silver is cleaned, deduplicated and conformed. Gold is shaped for a specific business use
Data product	A dataset treated like a product: it has an owner, a documented meaning, a quality guarantee, a freshness promise, and consumers who depend on it
Certified / curated / experimental	How much a dataset can be trusted. Certified is safe for executive reporting; experimental is not safe for anything that leaves the analyst's screen
Unity Catalog	The platform's single register of what data exists, who owns it and who may see it. One access model for every tool
Governed tag	A controlled label on data (for example, "contains personal data") that automatically triggers the right protection wherever it is applied
ABAC (attribute-based access control)	Access rules written against labels rather than against individual tables, so a correctly labelled new table is protected automatically
Pseudonymisation	Replacing a person's identifiers with a meaningless key, so analysis can happen without the analysis containing personal data
Right to erasure	The legal right of an individual to have their personal data deleted, and our obligation to prove we did it everywhere
Data residency	The requirement that certain data is stored, and sometimes processed, inside a particular country or region
Lineage	The automatically recorded map of where every number came from and what it feeds
Streaming vs batch	Streaming processes data as it arrives (seconds to minutes). Batch processes it on a schedule (hourly or daily). Streaming costs more and is only worth it when a decision is made that fast
Strangler migration	Replacing an old system by moving its users to a new one piece by piece and draining the old one, instead of a single switchover
Dual-run	A period during which old and new systems both produce the same numbers so they can be compared before the old one is switched off
Reconciliation	The automated daily comparison that proves the new number equals the old one, or explains why it should not
Decommission gate	A defined, evidenced condition that must be met before an old system is switched off
SLO (service level objective)	A published, measured promise — for example, "yesterday's sales are complete by 06:00 on 99 days out of 100"
Thin slice	A narrow but complete first delivery that goes through every layer, from source to a real user, so the whole approach is proven before it is scaled
OpenSharing (formerly Delta Sharing)	Giving an external party live, read-only access to a specific slice of data without sending them a copy or building them a portal
Metric view	A single, central definition of a business measure such as gross margin, used identically by every report and every tool
Feature table	A governed table of model inputs, so that a model is trained and run on identical definitions
DBU	The unit in which Databricks platform consumption is measured. Analogous to a metered utility unit

14.2 Assumptions register

Every assumption below is illustrative, is load-bearing somewhere, and must be tested. The "test in" column names the phase in which it stops being an assumption.

#	Assumption	Affects	Test in	If wrong
A1	Client is fictional; all figures illustrative	Everything	—	—
A2	2,400 stores, 14 countries, e-commerce in 9, $34bn revenue, 40m loyalty members, 85,000 employees	Sizing, value case	Phase 0	Re-base entirely
A3	~4.0m baskets and ~23m POS line items per day	Ingestion sizing, compute	Phase 0	Linear effect on ingestion cost
A4	~360m clickstream events/day, 22k/sec peak, 3x Black Friday	Streaming design, compute	Phase 0	Streaming architecture and cost scale directly
A5	~40m changed SAP rows/day, 4-hour inventory latency requirement	SAP extraction decision (D-03)	Phase 0	A looser latency requirement permits a cheaper extraction route
A6	Log-based CDC on the SAP source database is obtainable	D-03, Phase 2 plan	Phase 0	Fall back to ODP/SLT; add 4–6 weeks and client-side licence cost
A7	Identity match rate between store loyalty and e-commerce reaches 70–80%	UC-1, UC-6 value	Phase 0 (sample)	Below 55%, UC-6 value roughly halves and should be resequenced
A8	Book stock accuracy ~88%, improvable to 95%+	UC-3	Phase 2	UC-3 will be rejected by stores; a stock-accuracy programme becomes a prerequisite
A9	Forecast accuracy improves from ~38% to 26–30% WMAPE	UC-2 value	Phase 2 pilot	Inventory benefit scales roughly with the improvement delivered
A10	Supply Chain re-cuts safety-stock parameters on the improved forecast	UC-2 value realisation	Phase 3	The largest single benefit does not land. **This is an operating-model risk, not a data risk**
A11	~620 active EDW reports, of which 35–45% can be retired	Migration effort (§8.5)	Phase 2	Each 10 percentage points of retirement shortfall adds roughly 8–10 weeks of migration effort
A12	60–90 TB of EDW history is worth retaining	Storage, migration	Phase 0	Storage and export effort scale directly
A13	Hadoop estate is ~60% cold and largely not worth migrating	Migration scope	Phase 0	Migrating more adds effort and permanent storage cost
A14	Kafka retention can be raised to 7 days and a schema registry adopted	Clickstream reliability	Phase 1	Reliability capped; incident rate stays high; rescued-data volumes grow
A15	400+ analysts, bursty query pattern	D-07, SQL cost	Phase 0 (from legacy query logs)	A steady, predictable pattern would favour provisioned compute for part of the load
A16	Residency requires in-region processing in EU, UK and 3 APAC jurisdictions	D-08, workspace count	Phase 0 (Legal)	Fewer regions materially simplifies and reduces cost; more regions does the reverse
A17	Works-council consultation gates worker-level HR data in 5+ countries	§3.10, scope	Phase 0	People domain resequences
A18	~1,200 suppliers can consume a data share, with a thin UI for the smallest	D-14	Phase 2 pilot (20 suppliers)	A light portal becomes a funded workstream
A19	12–16 of the existing 48-person team can transition into platform roles	Handover, §9.3	Phase 2	Handover slips; either hire or extend the services engagement
A20	Dual-run overlap runs weeks 41–88 (47 weeks); Phase 4 itself is 24 weeks	Cost peak, §10.4	Phase 3	Every additional quarter of dual-run is a material, visible cost
A21	Inventory ~$3.4bn, carrying cost ~18%, gross margin ~28%, markdown ~5.6% of revenue	Value case arithmetic	Phase 0 (Finance)	Benefit ranges scale directly
A22	Benefit capture at 60% of the identified low case, delayed two quarters	§10.6 payback	Continuously	Payback moves, but remains inside Year 2 down to roughly 40% capture

14.3 Items marked to be confirmed

Item	Where	Confirm with
SAP third-party extraction licensing implications	§3.1	Client's SAP account team
Per-object incremental support in the Salesforce managed connector	§3.2	Current connector documentation
Whether a managed streaming connector now covers the clickstream requirement	§5.3	Current documentation, at the Phase 3 gate
Current ABAC compute requirements and policy quotas	§5.5, D-06	Current documentation, at build time
Low-latency operational serving pattern and product choice	§5.6	Current documentation

Item	Where	Confirm with
Cross-metastore access options for multi-region deployments	§5.7, D-08	Current documentation, at design time
Audit log retention and any configurable retention options	§7.7	Current documentation
Cross-region and cross-cloud sharing transfer costs for non-EU suppliers	§7.6, D-14	Current documentation and cloud pricing
Food-safety record-keeping retention periods per country	§5.2, §7.1	Client Legal
Disaster recovery RPO and RTO targets per zone	§9.6	Client
All platform consumption pricing	§10	Current price lists, with the Databricks account team and the cloud provider, in Phase 0

End of case 01. Prepared as a teaching and selling artefact. The client is fictional; the figures are illustrative; the architecture decisions are real positions and are meant to be argued with.

End of document. Meridian Retail Group — Global Omnichannel Lakehouse · Solution design — technical annex · 27 September 2026 · Illustrative — fictional client; all figures illustrative.